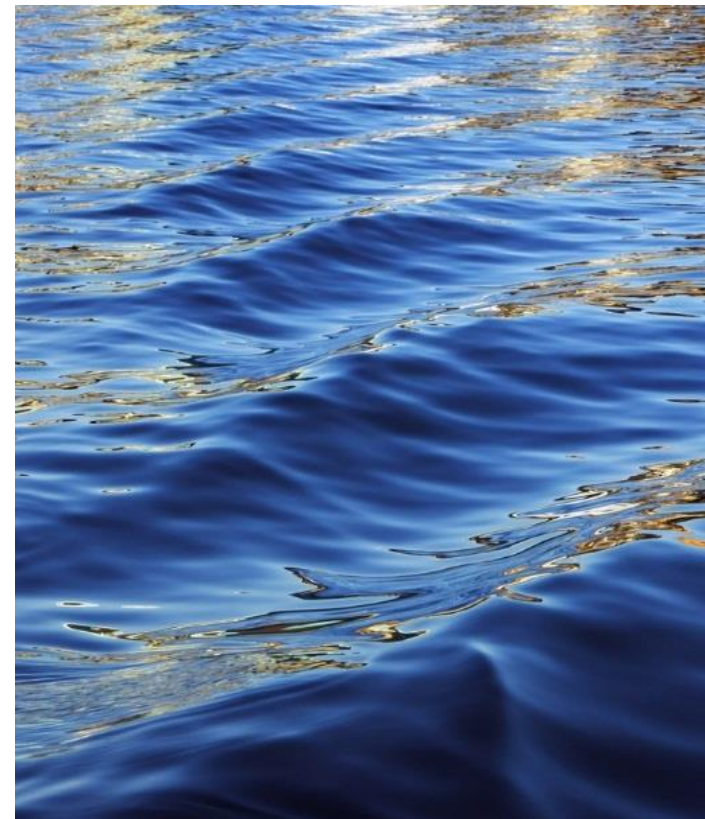




Lecture 10

Morphological Image Processing

Digital Image Processing



What we know for now & What we going to do?

It is impossible of course to design a filter that removes any noise but keeps all the important image structures intact because no filter can discriminate which image content is important to the viewer and which is not.

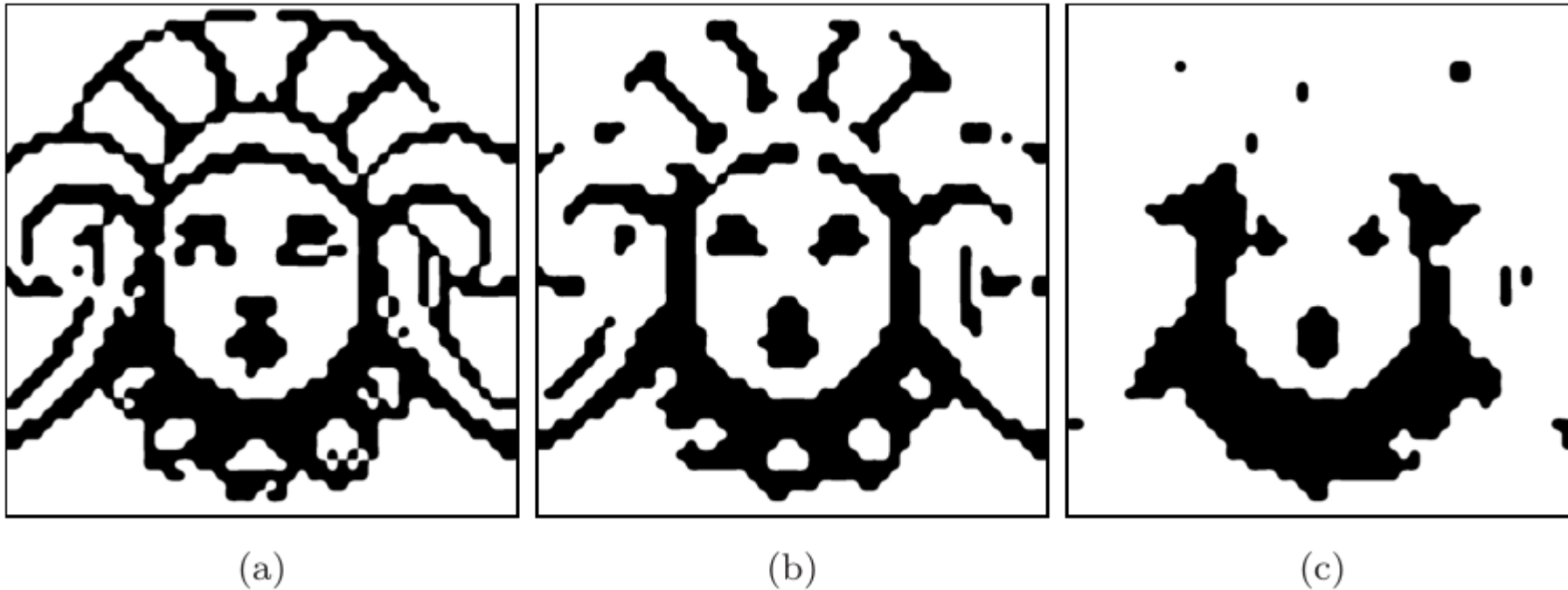
Let's say if we try to apply a "Median filter" to remove noise.

This operation will modify the intensity in a statistic theme, but we can't control structure of it.



See example in next slide

What we know for now & What we going to do?



An example of median filter result.

We can't control which black or white pixels, need to
grow or **shrink** (we hardly control it shape).

What we know for now & What we going to do?

So, that why we are going to focus on "Morphological filters" in this lecture.

In the beginning, we are just focus on a **binary image** with contain only two possible values, 0 and 1 for black and white.

NOTE:

Generally, we denote binary with value 1 as a "foreground" and value 0 as a "background" pixels.

The basic theory that we need to know before...

"Set theory"

Definition in mathematic form

- ถ้า w เป็นเป็น Pixel ในภาพ เราจะเขียนได้ว่า $w = (x, y)$
- ถ้า w เป็น Element ของ A เราจะเขียนได้ว่า $w \in A$
- ถ้า w ไม่เป็น Element ของ A เราจะเขียนได้ว่า $w \notin A$
- ถ้าทุก Element ของ A เป็นสมาชิกของ B ทั้งหมด $A \subseteq B$ "Subset"
- ถ้า Set ของ A ไม่มี Element อยู่เลย $A = \{\emptyset\}$ "Empty set"
(Empty set) หรือ (Null)

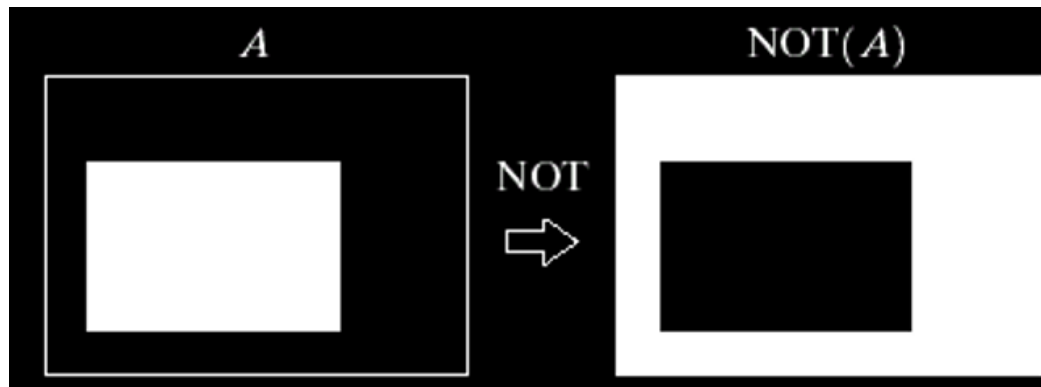
The basic theory that we need to know before...

"Set theory"

Definition in mathematic form

□ การเขียน Condition ของ Set

$$A = \{w | \text{condition}\}$$

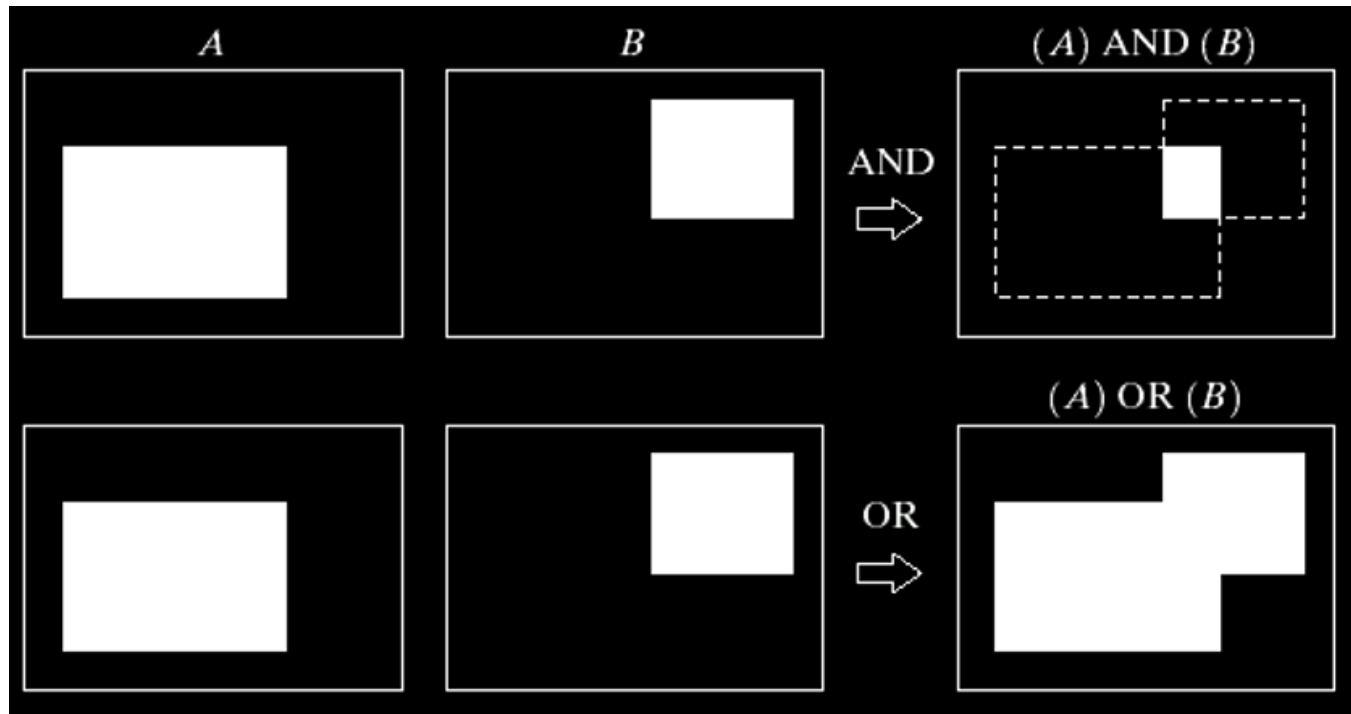


Complement

$$A^c = \{w | w \notin A\}$$

The basic theory that we need to know before...

"Set theory"



Intersection

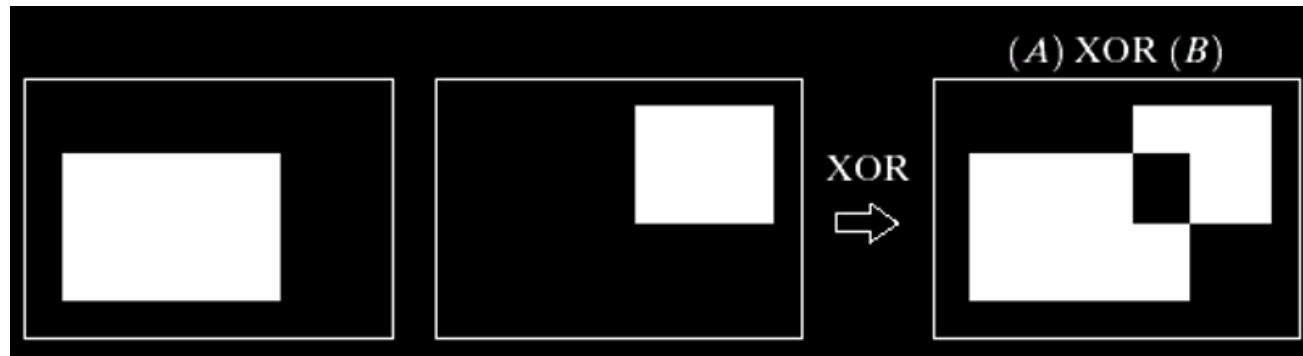
$$C = A \cap B$$

Union

$$C = A \cup B$$

The basic theory that we need to know before...

"Set theory"



Exclusive or

$$C = (A \cup B) - (A \cap B)$$



Difference

$$C = B - A$$

$$C = B \cap A^c$$

$$C = \{w | w \in B, w \notin A\}$$

The basic theory that we need to know before...

"Set theory"

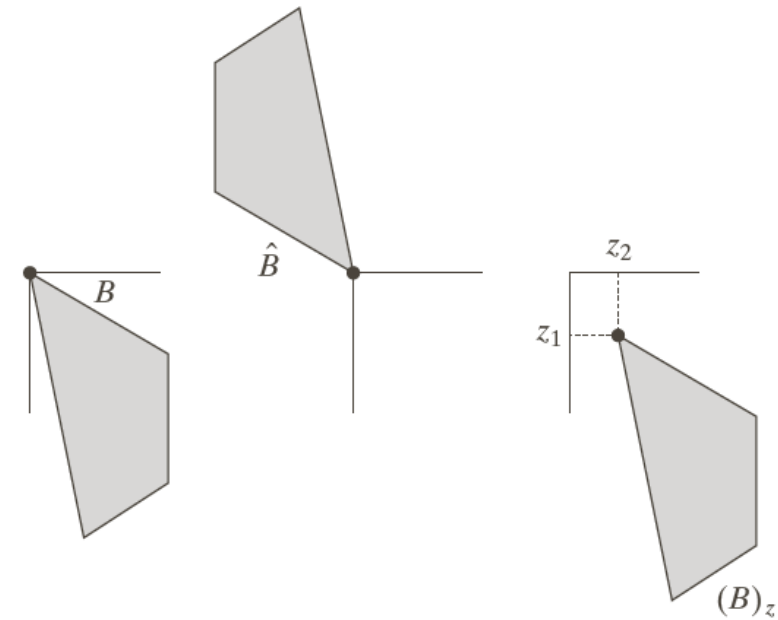
Reflection

$$\hat{B} = \{w | w = -b, \quad \text{for } b \in B\}$$

Translation (Displacement)

translate by point $z = (z_1, z_2)$

$$(B)_z = \{w | w = b + z, \quad \text{for } b \in B\}$$



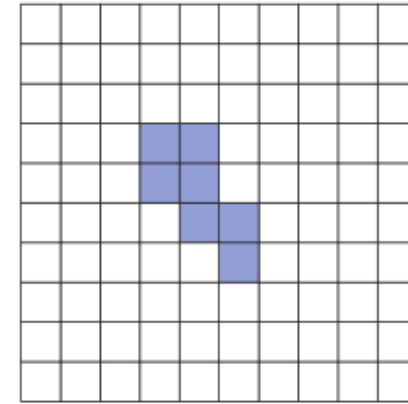
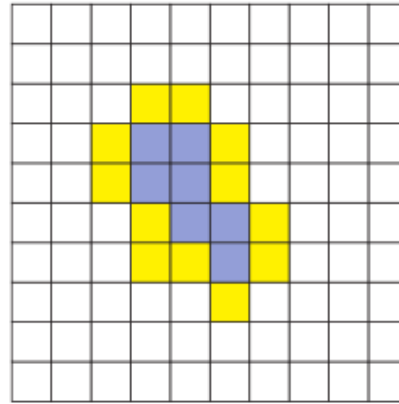
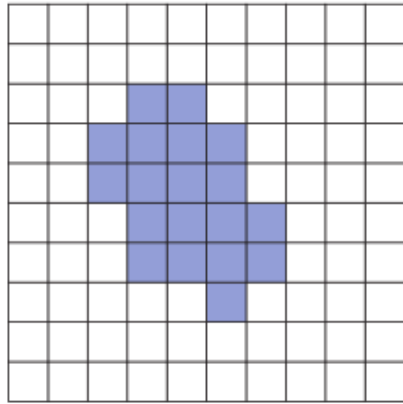
The dot identifies the
origin of the sets.



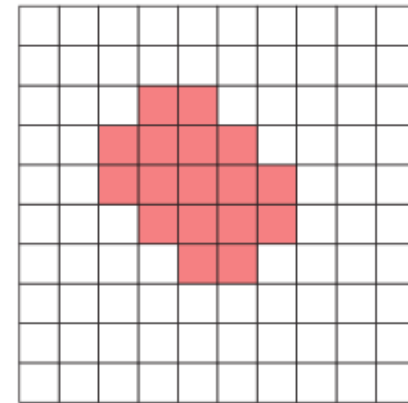
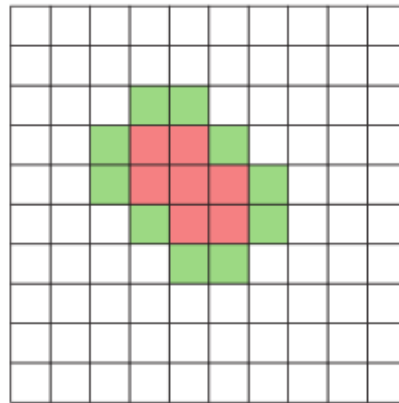
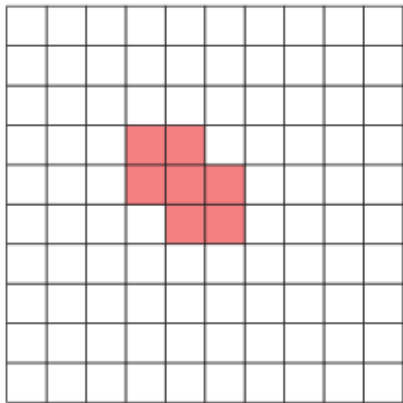
How can we control the size and possibly the shape of the structures affected by filtering operation?

The basic idea is "Shrink" & "Grow"





“Shrinking” a foreground region by removing a layer of border pixels



“Growing” a foreground region by attaching a layer of pixels

The fundamental operator of morphological filtering

“Dilation” is an operation that “grows” or “thickens” objects in a binary image

“Erosion” is an operation that “shrinks” or “thins” objects in a binary image.

The specific manner and extent of these operators are controlled by a shape referred to as a **“structuring element”**.

ตัวอย่างเช่น ถ้าอยากให้วัตถุออกไปด้านซ้าย - ขวาเท่านั้น ไม่ยืดบน - ล่าง
เราต้องปรับที่รูปร่างของ Structuring element

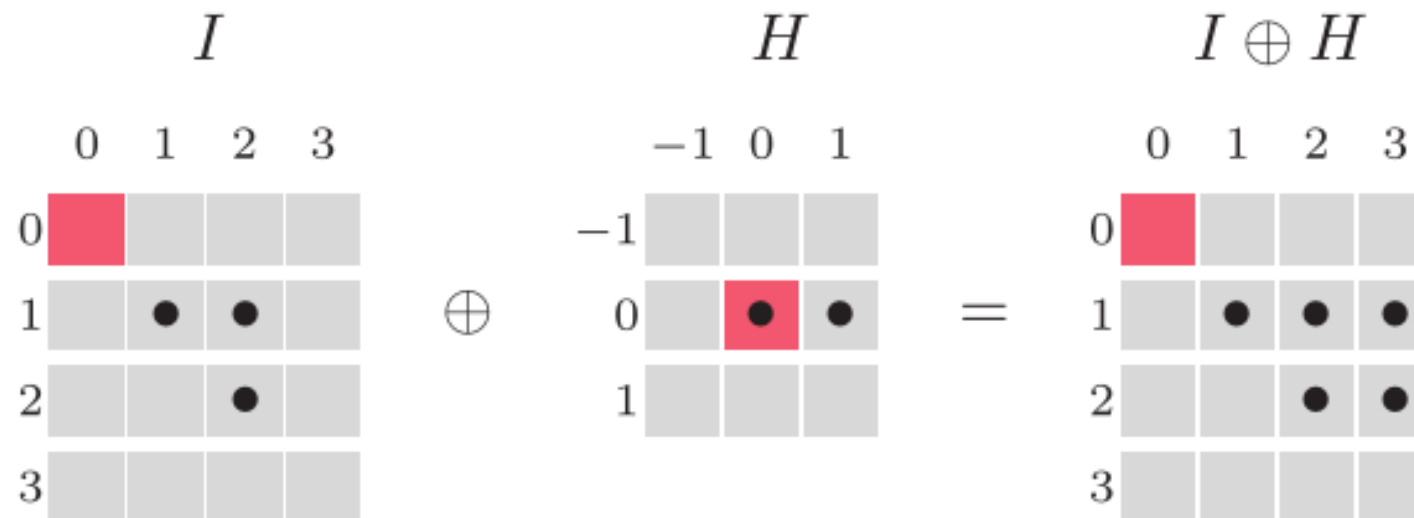
Dilation

The dilation of I and H is denote by

$$I \oplus H = \{z | (\hat{H})_z \cap I \neq \emptyset\}$$

z : set of all coordinates in the image (for displacement)

H : refer as “Structuring element” in dilation



“Red block” refer as a center of each object.

Dilation : Properties

Commutative

$$A \oplus B = B \oplus A$$

Associative

$$A \oplus (B \oplus C) = (A \oplus B) \oplus C$$

Note on decomposed

$$\text{If } B = B_1 \oplus B_2$$

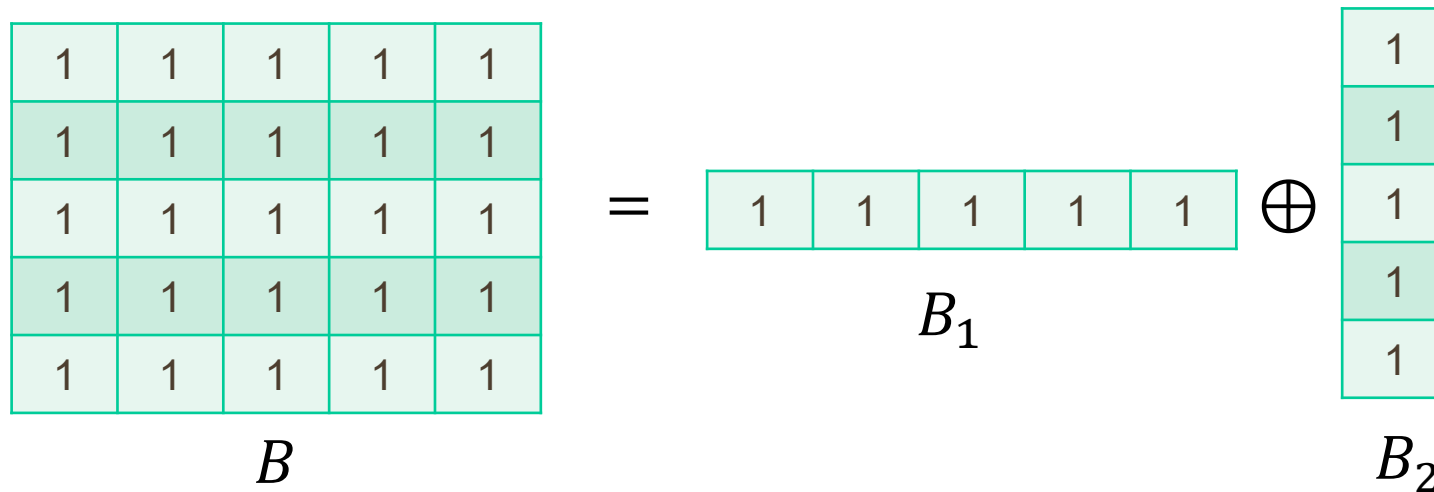
$$\text{Then } A \oplus B = A \oplus (B_1 \oplus B_2) = (A \oplus B_1) \oplus B_2$$

Dilation : Properties

Note on decomposed (continue from previous slide)

$$\text{If } B = B_1 \oplus B_2$$

$$\text{Then } A \oplus B = A \oplus (B_1 \oplus B_2) = (A \oplus B_1) \oplus B_2$$



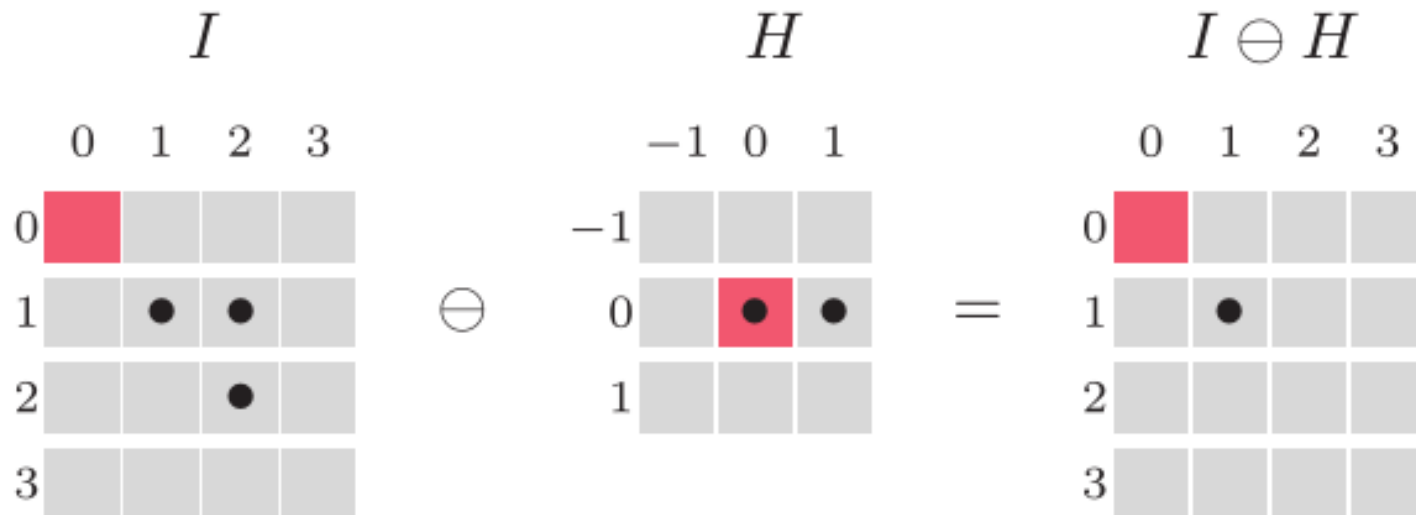
Erosion

The erosion of I and H is denote by

$$I \ominus H = \{z | (H)_z \subseteq I\}$$

z : set of all coordinates in the image (for displacement)

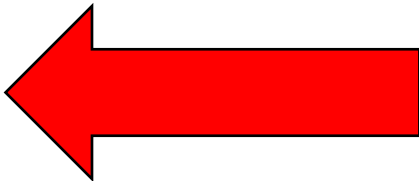
H : refer as “Structuring element” in erosion



“Red block” refer as a center of each object.

Erosion : Properties (need to be consider)

Be careful, some property not the same as dilation

$$I \ominus H \neq H \ominus I$$


Dilation & Erosion are **duals of each other**

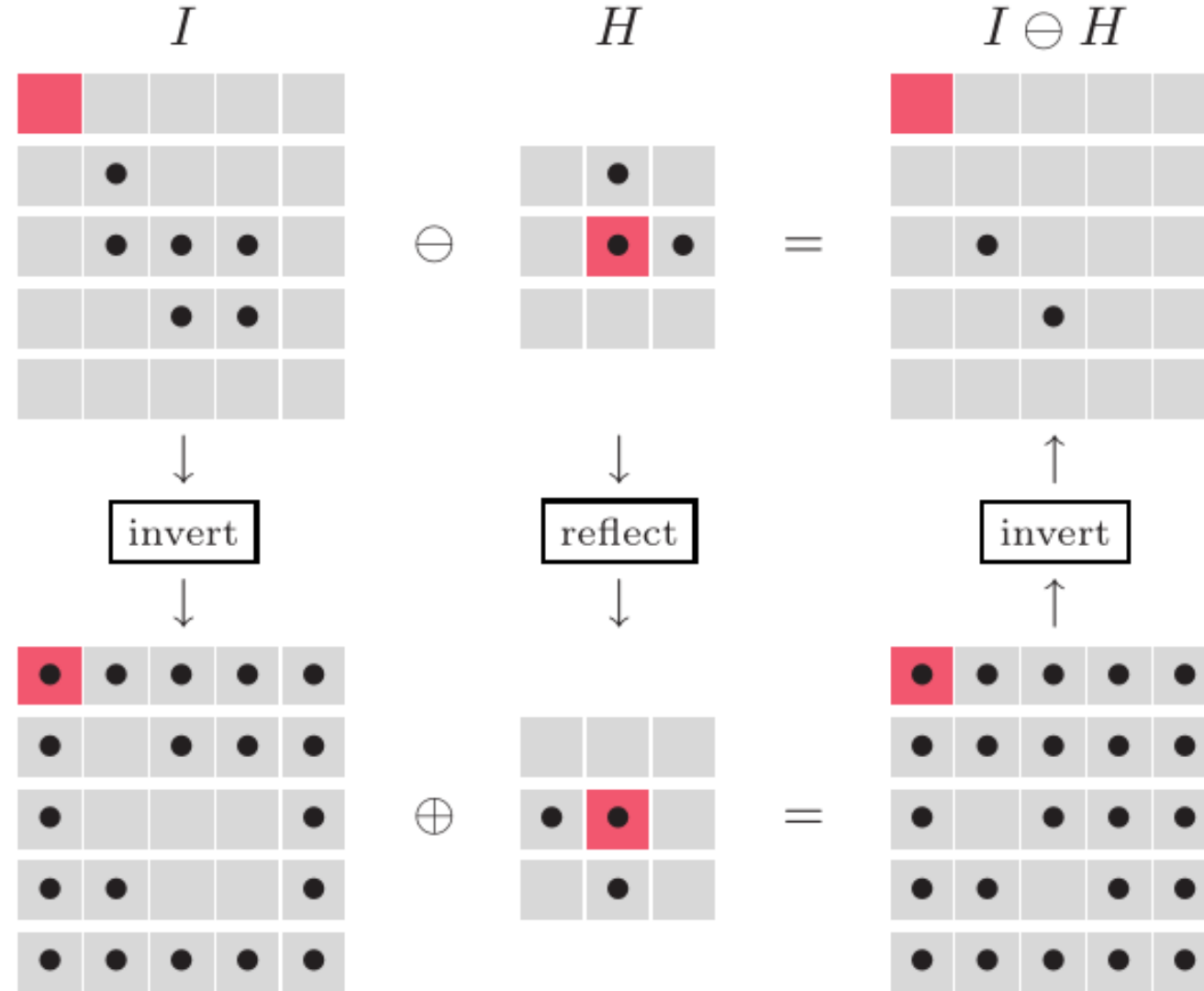
$$(A \ominus B)^c = A^c \oplus \hat{B}$$

“Dilation” and “Erosion”
are not mutually inverse

Chain rule on Dilation & Erosion

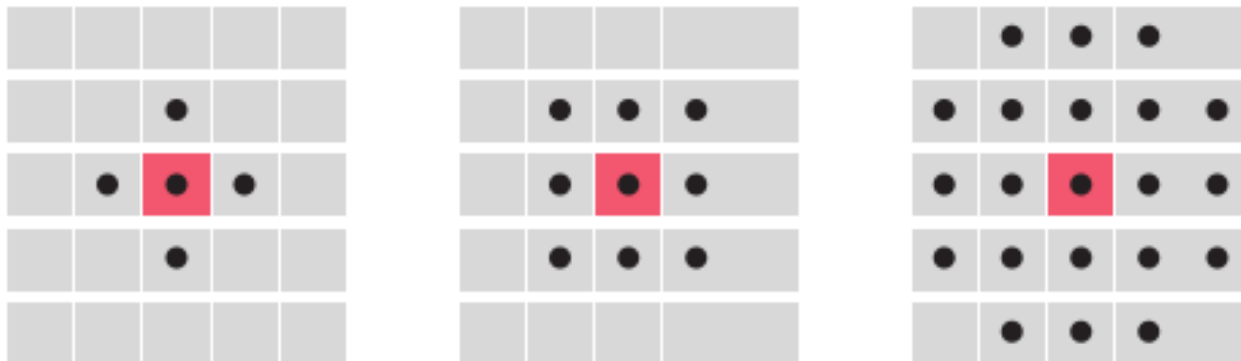
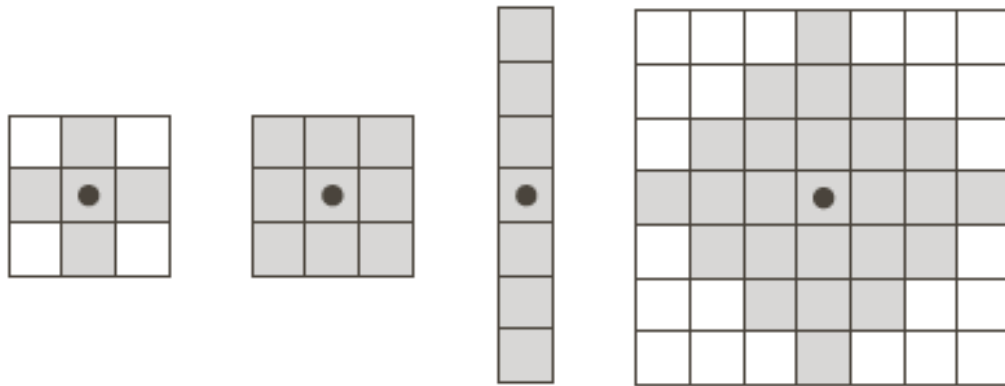
$$(A \ominus B) \ominus C = A \ominus (B \oplus C)$$

Dilation & Erosion are duals of each other

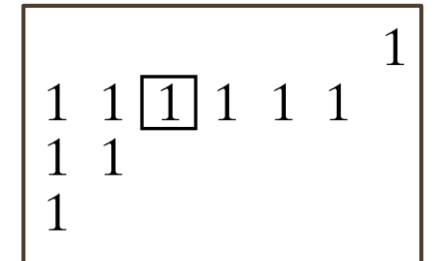
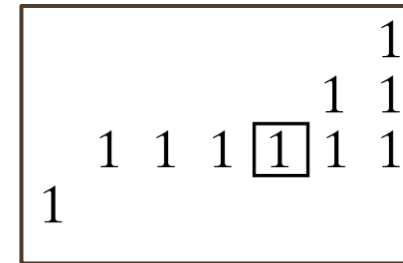


How to create a structuring element?

Symmetric structuring element



Non-symmetric structuring element



Sometime, origin is not at the center...

Python

```
1 import cv2 as cv
2 import numpy as np
3
4 img = cv.imread('sample.png', 0)
5 kernel = cv.getStructuringElement(cv.MORPH_ELLIPSE,(5,5))
6
7 erosion = cv.erode(img,kernel,iterations = 1)
8
9 dilation = cv.dilate(img,kernel,iterations = 1)
10
11 cv.imwrite('output_erosion.png', erosion)
12 cv.imwrite('output_dilation.png', dilation)
13
```

https://docs.opencv.org/3.4/d9/d61/tutorial_py_morphological_ops.html

What we know for now & What we going to do?

Now, we know how to make an object grow and shrink with specify shape (dilation & erosion). This lecture going to give you the idea to make use of these simple to make some useful application.

Firstly, we are going to learn about something more complicate which call: **Opening** & **Closing** operation.



Opening & Closing Morphological Processing

Opening

A binary opening $I \circ H$ denotes an erosion followed by a dilation with the *same* structuring element H ,

$$I \circ H = (I \ominus H) \oplus H.$$

Closing

When the sequence of erosion and dilation is reversed, the resulting operation is called a closing and denoted $I \bullet H$,

$$I \bullet H = (I \oplus H) \ominus H.$$

Opening & Closing Morphological Processing

Morphological opening removes completely regions of an object that cannot contain the structuring element, smooth object contours, breaks thin connections, and removes thin protrusions.

Erosion

>>>>>

Dilation

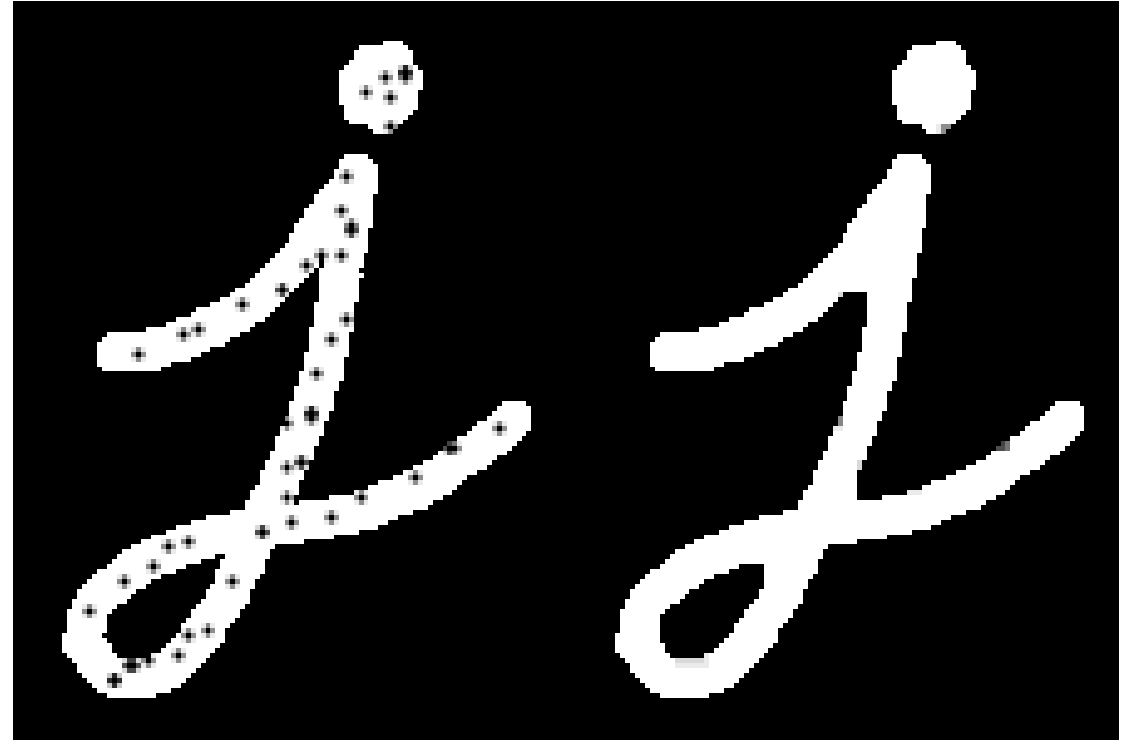
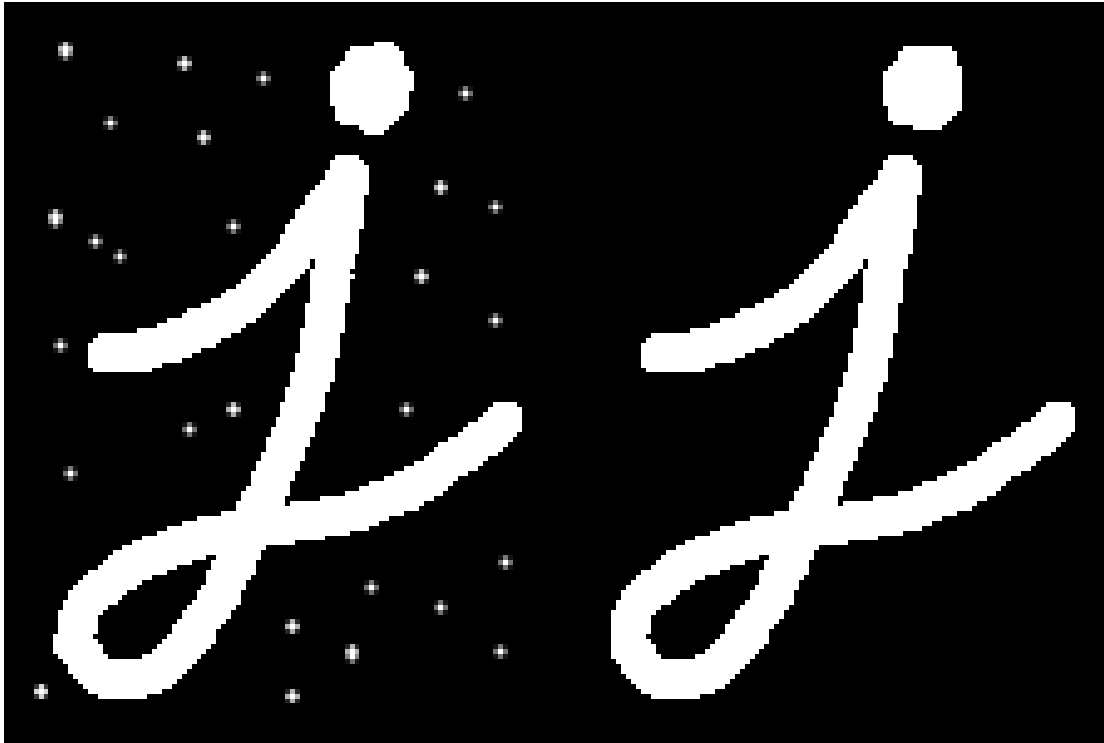
Morphological closing tends to smooth the contours of objects. Unlike opening, it generally joins narrow breaks, fills long thin gulfs, and fills holes smaller than the structuring element.

Dilation

>>>>>

Erosion

Opening & Closing Morphological Processing

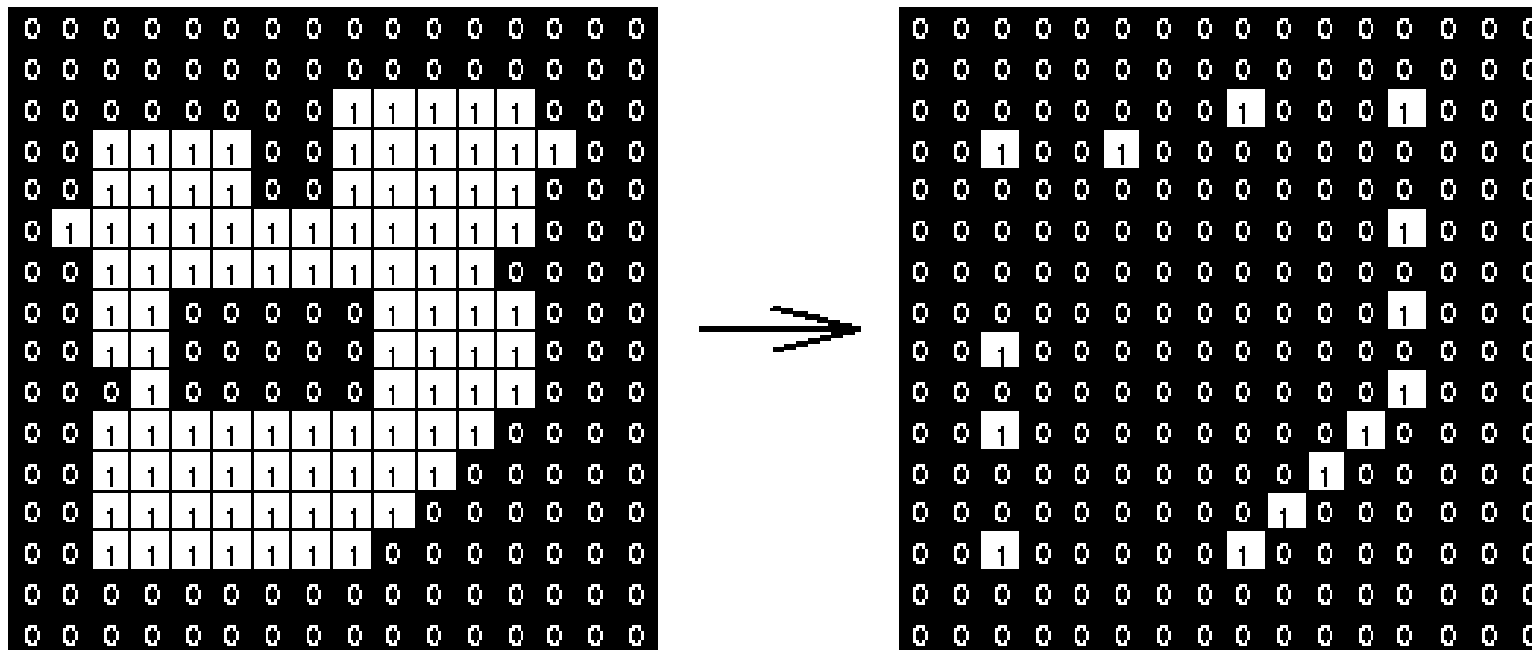


Do you know which one is **opening** / **closing**?

*** Please note that the answer depend on the **shape of SE (white / black)**

Hit-or-Miss Morphological Processing

It is useful to be able to identify specified configurations of pixels, such as isolated foreground pixels, or pixels that are end points of line segments.



Hit-or-Miss Morphological Processing

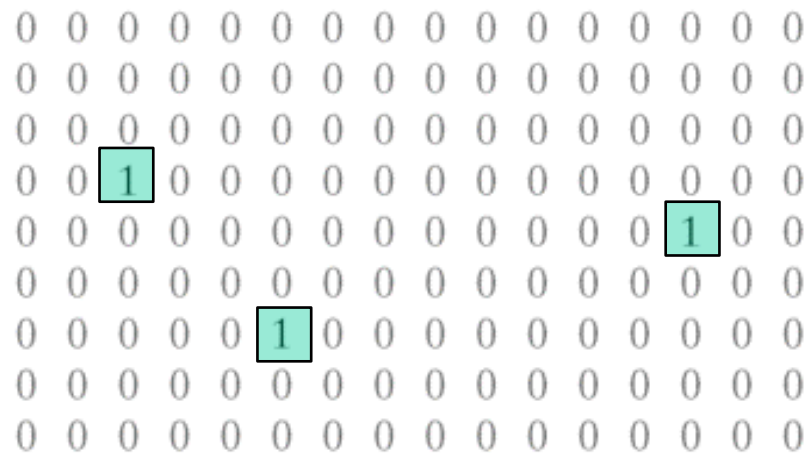
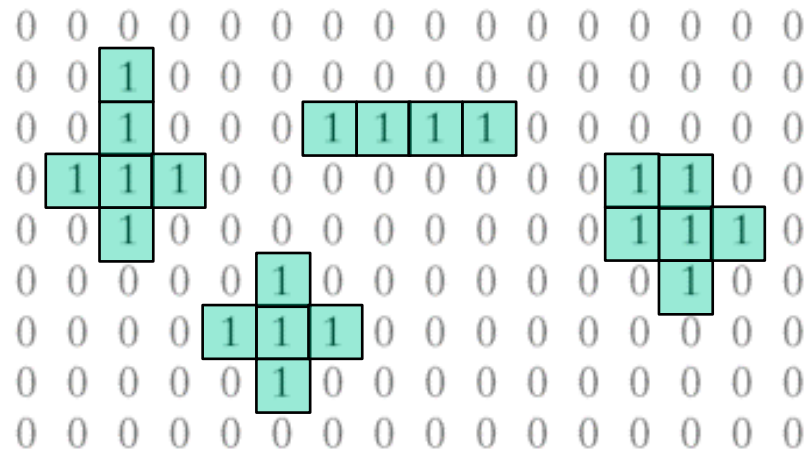
The Hit-or-Miss transform of A by B is denoted by

$$A \odot B = (A \ominus B_1) \cap (A^c \ominus B_2)$$

The name "hit-or-miss transformation" is based on how the result is affected by the two erosions. For example, the output image consists of all locations that match the pixels in B_1 (a "hit") and that have none of the pixels in B_2 (a "miss").

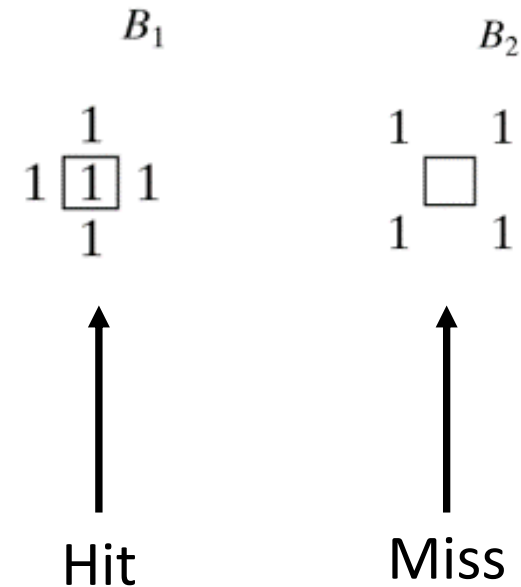
The accurate name should be "hit-and-miss transformation".

Hit-or-Miss Morphological Processing



$(A \ominus B_1)$

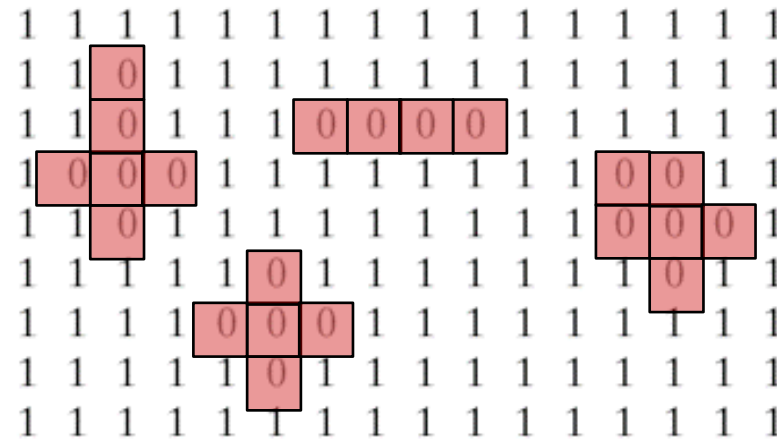
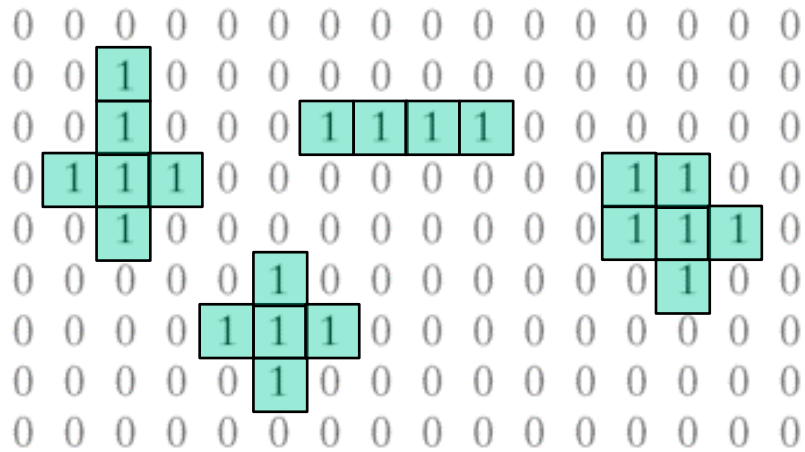
$$A \circledast B = (A \ominus B_1) \cap (A^c \ominus B_2)$$



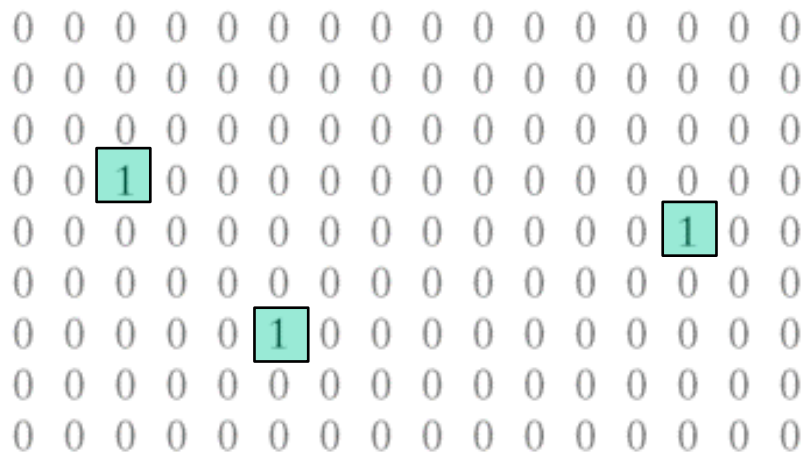
Let's say, if we would like to find a "plus" shape in the image

Hit-or-Miss Morphological Processing

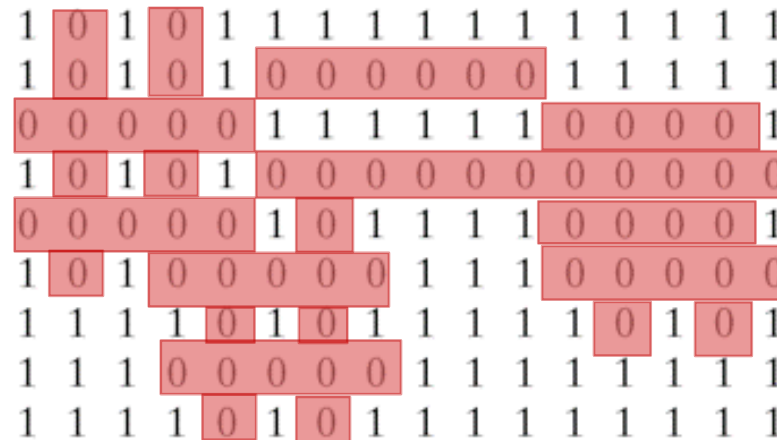
$$A \odot B = \underline{(A \ominus B_1)} \cap \underline{(A^c \ominus B_2)}$$



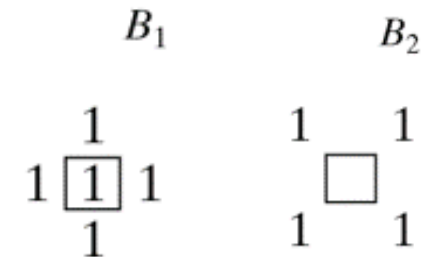
A^c



$(A \ominus B_1)$

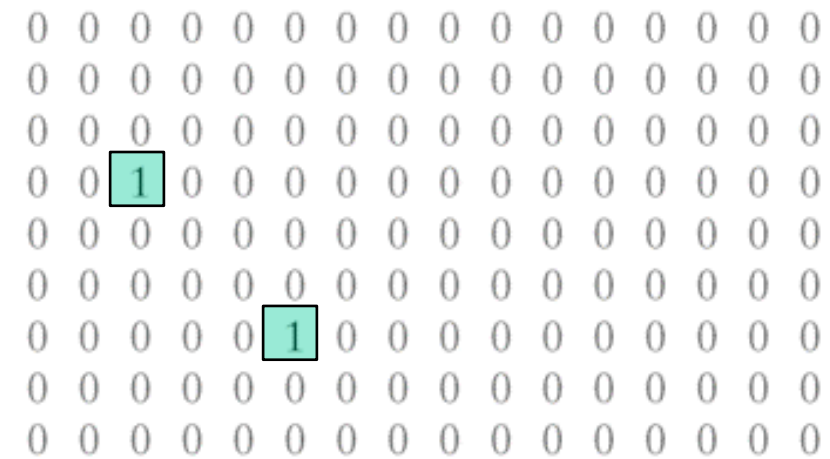
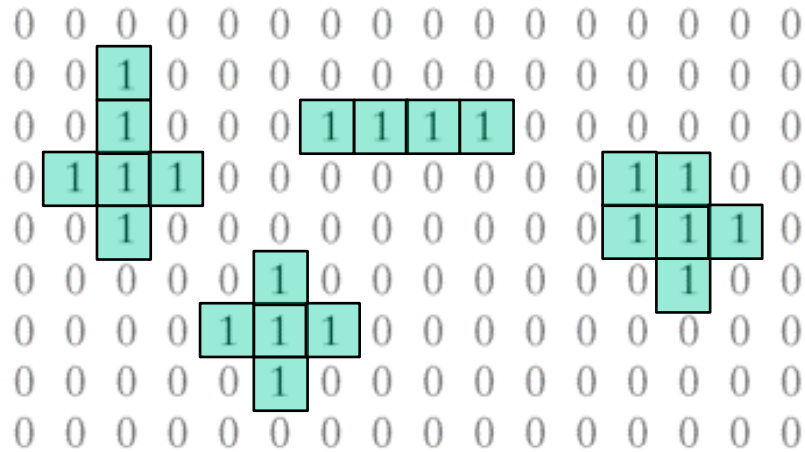


$A^c \ominus B_2$



Hit-or-Miss Morphological Processing

$$A \odot B = (A \ominus B_1) \cap (A^c \ominus B_2)$$



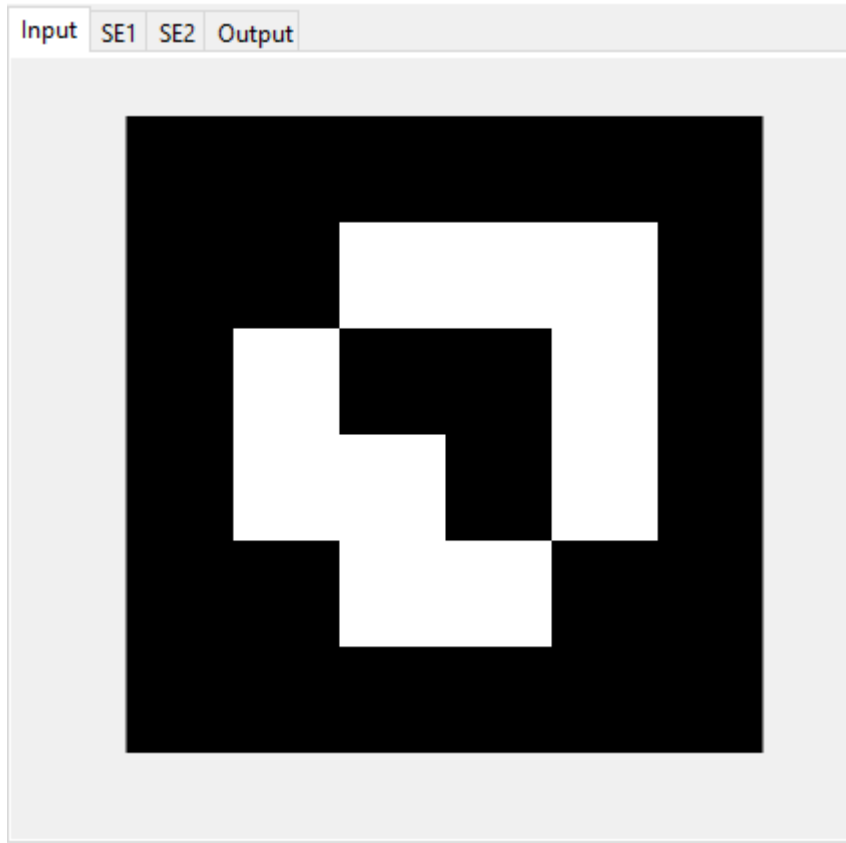
$$B_1$$

$$\begin{array}{ccc} & 1 & \\ 1 & \boxed{1} & 1 \\ & 1 & \end{array}$$

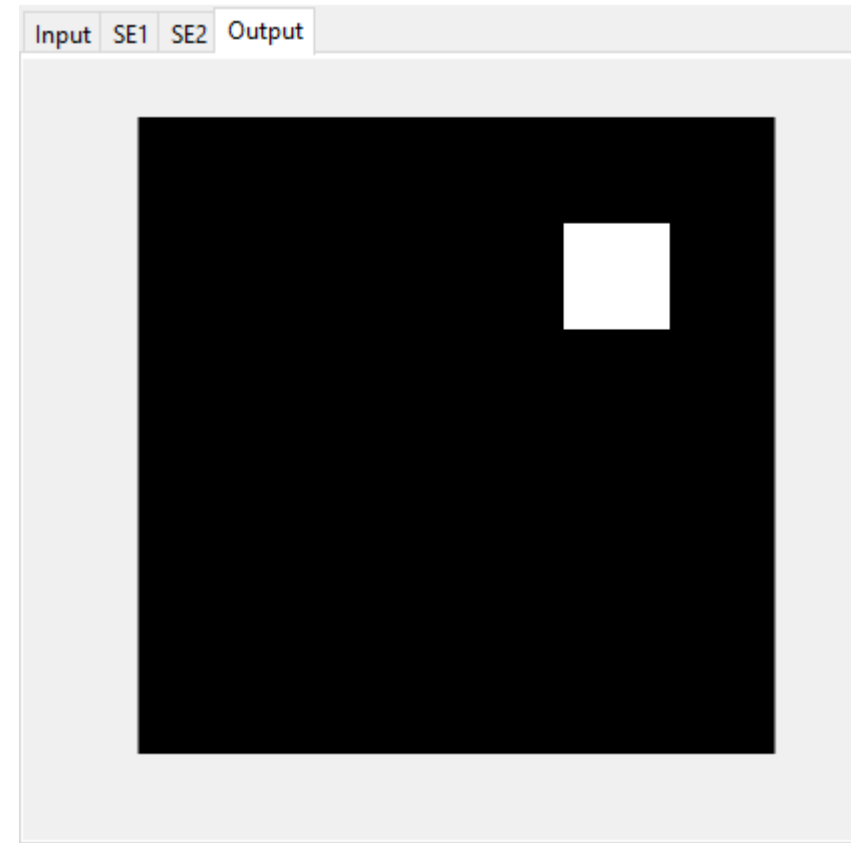
$$B_2$$

$$\begin{array}{ccc} 1 & & 1 \\ & \square & \\ 1 & & 1 \end{array}$$

Hit-or-Miss Morphological Processing

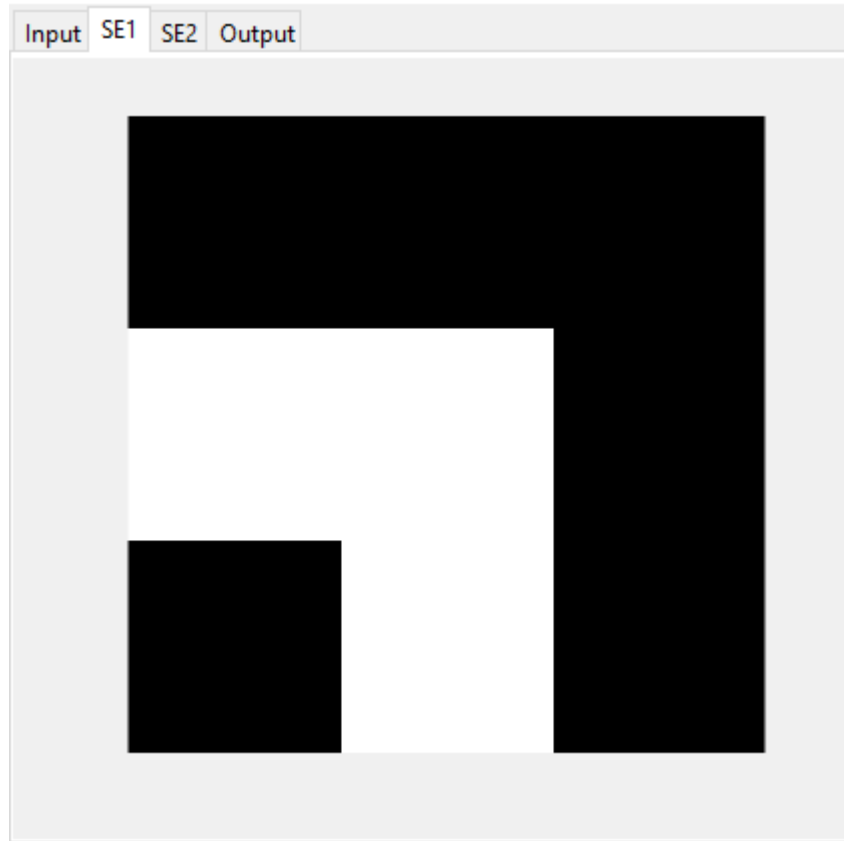


Input

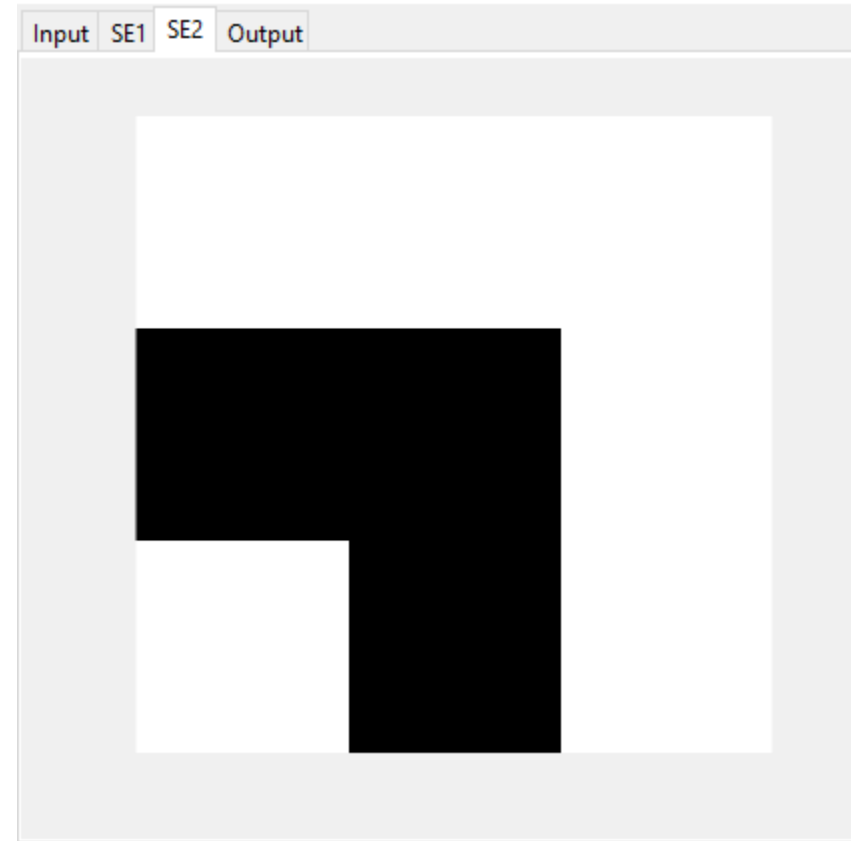


Output

Hit-or-Miss Morphological Processing



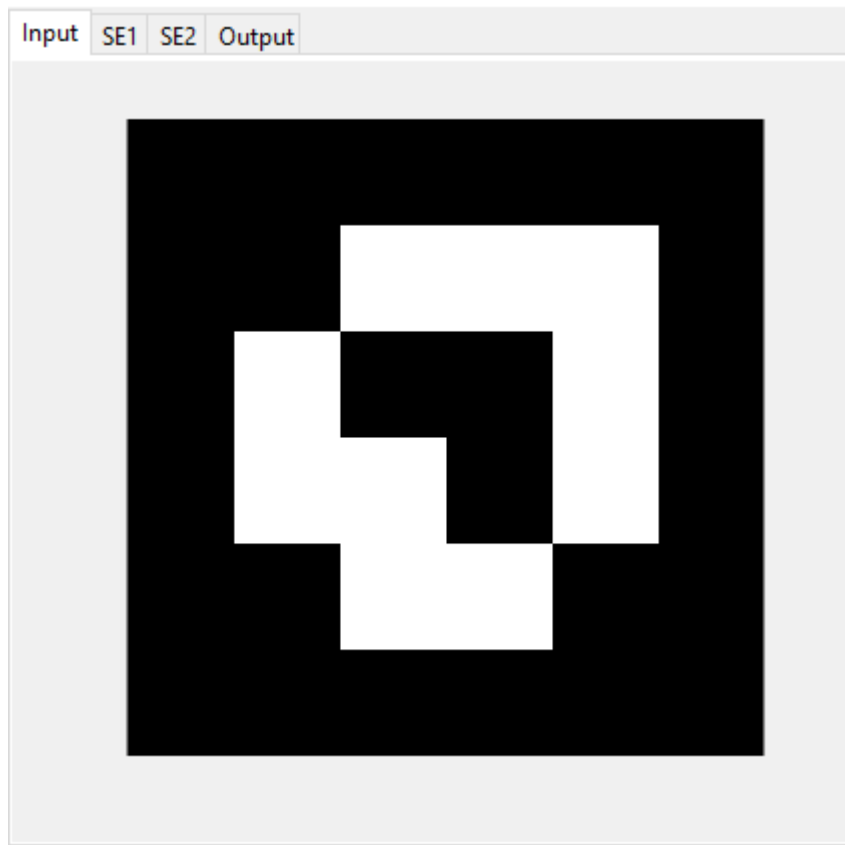
SE1



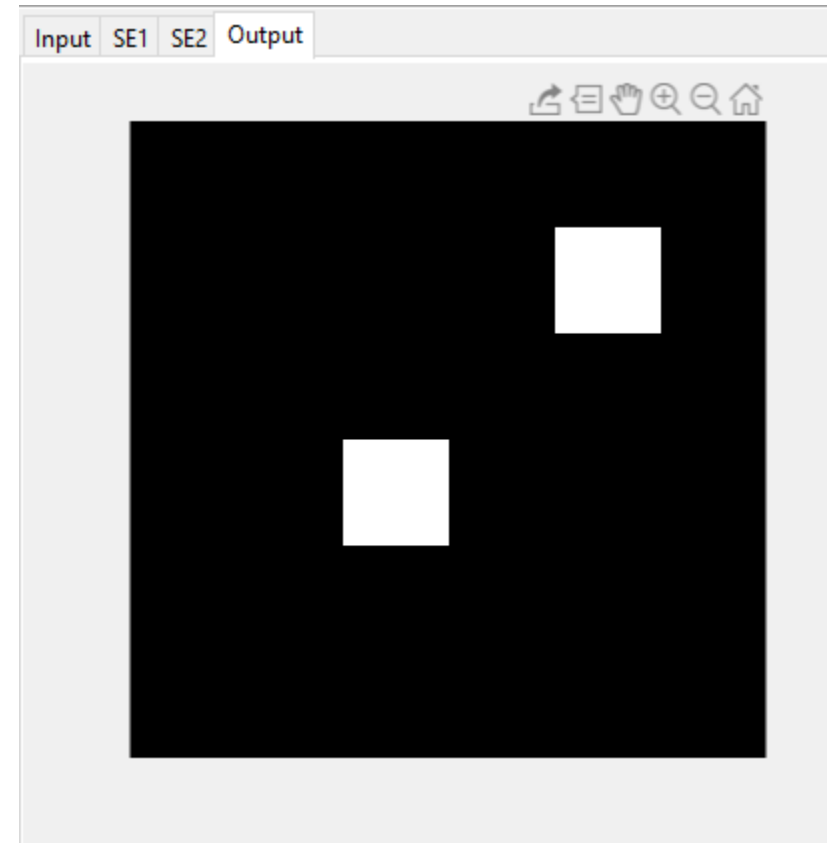
SE2

Hit-or-Miss Morphological Processing

Quiz
SE shape ?



Input



Output

Another useful morphological operation :)

- Boundary Extraction

- Region Filing

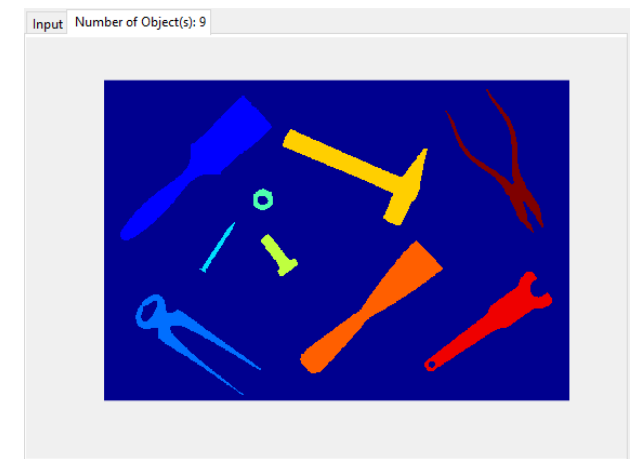
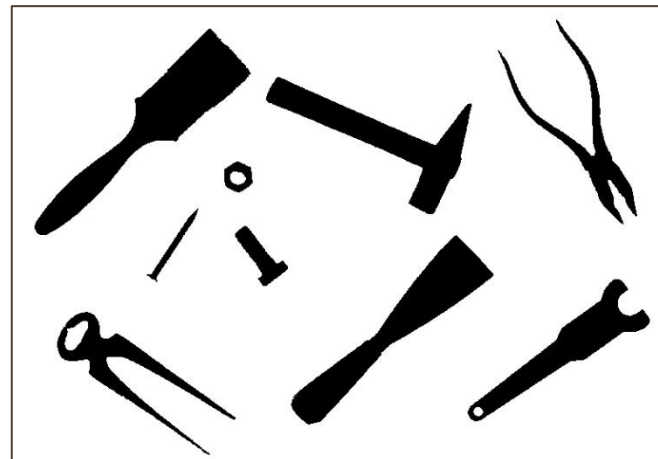
- Convex Hull

- Thinning

- Thickening

- Skeletons

- Region Labeling

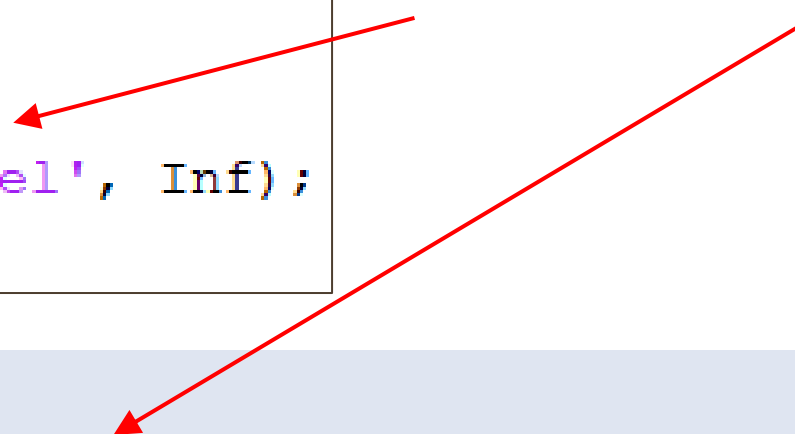


Implementation with MATLAB and Python

MATLAB

```
6
7 - in = imread('tools_w.jpg');
8 - if size(in,3)>1
9 -     in = rgb2gray(in);
10 - end
11 - in = imbinarize(in);
12
13 - out = bwmorph(in, 'skel', Inf);
14
```

Choose morphology operation and parameter



Python:

```
dst = cv.morphologyEx( src, op, kernel[, dst[, anchor[, iterations[, borderType[, borderValue]]]] ] )
```

Region filling ของ Python แนะนำให้ตามไปอ่านใน Link นี้แทน เนื่องจากไม่มี Morphology Operation ให้เลือกใช้
<https://learnopencv.com/filling-holes-in-an-image-using-opencv-python-c/>

Morphological processing (Binary image → Grayscale image)

Now, we use morphology only for binary image. However, it also can be applying in grayscale image.

In fact, the definition of grayscale morphology is a generalization of binary morphology, with the binary **OR** and **AND** operators replaced by the arithmetic **MAX** and **MIN** operators, respectively.

Please note that: In the case of colour images, the grayscale operations are usually applied individually to each colour channel.

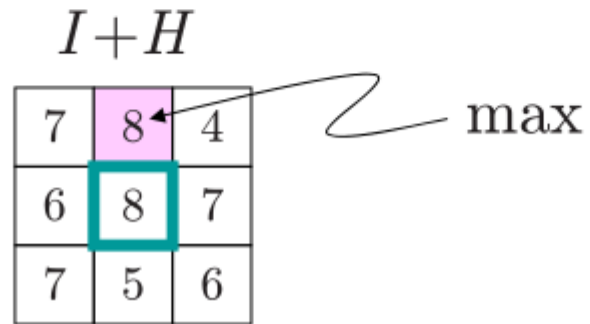
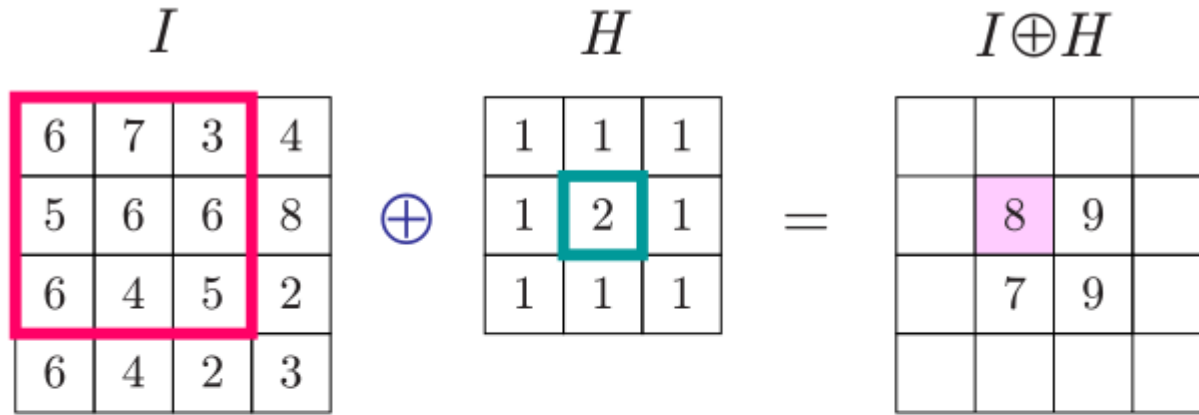
Structure Element (SE) in grayscale morphology

Unlike in the binary scheme, the structuring elements for grayscale morphology are not defined as point sets but as real-valued 2D functions,

The design of structuring elements for grayscale morphology must therefore distinguish explicitly between cells containing the value 0 and empty ("don't care") cells; for example

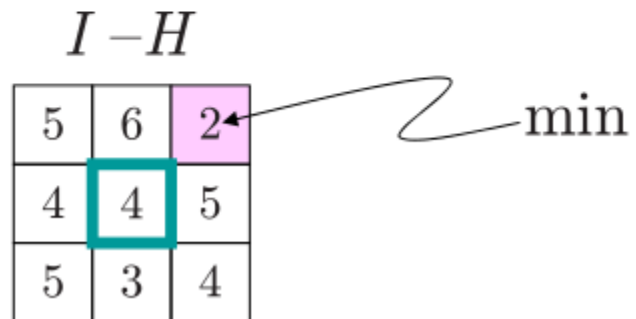
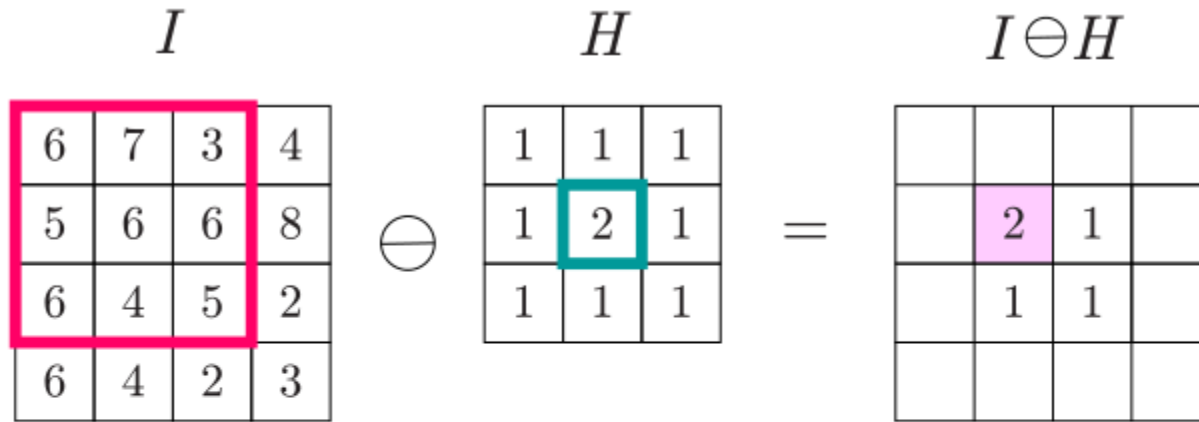


Dilation in grayscale morphology



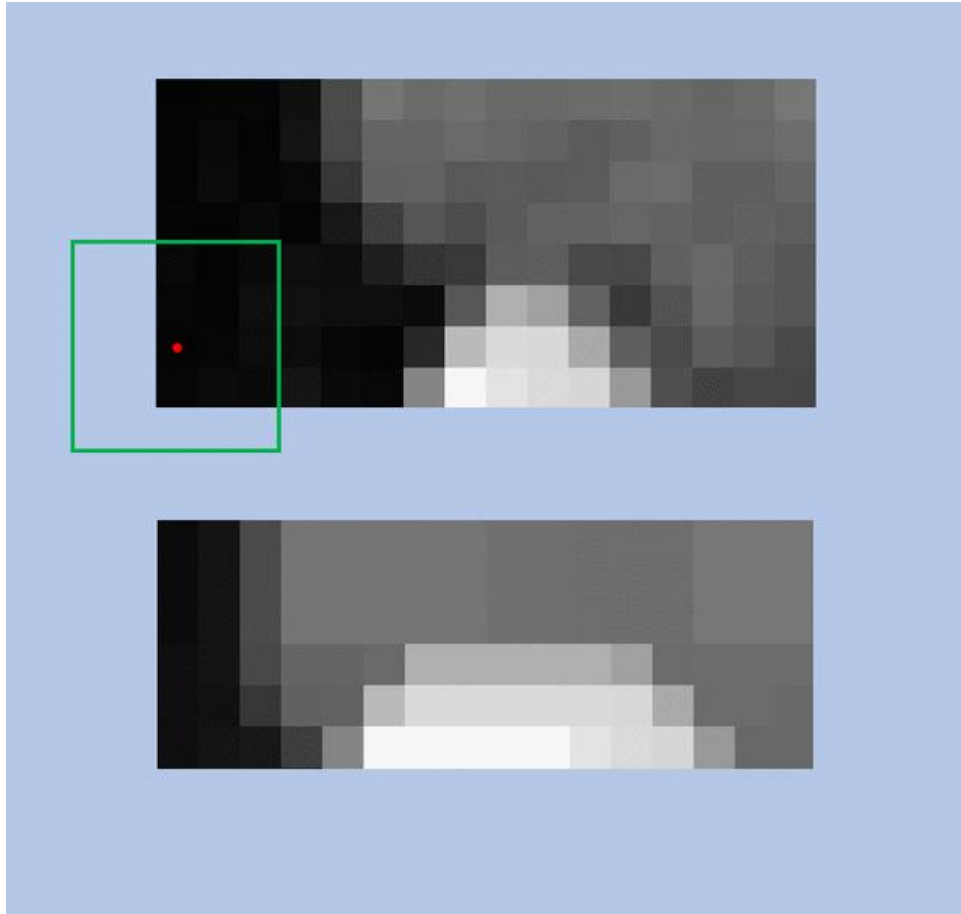
$$(I \oplus H)(u, v) = \max_{(i,j) \in H} \{I(u+i, v+j) + H(i, j)\}$$

Erosion in grayscale morphology

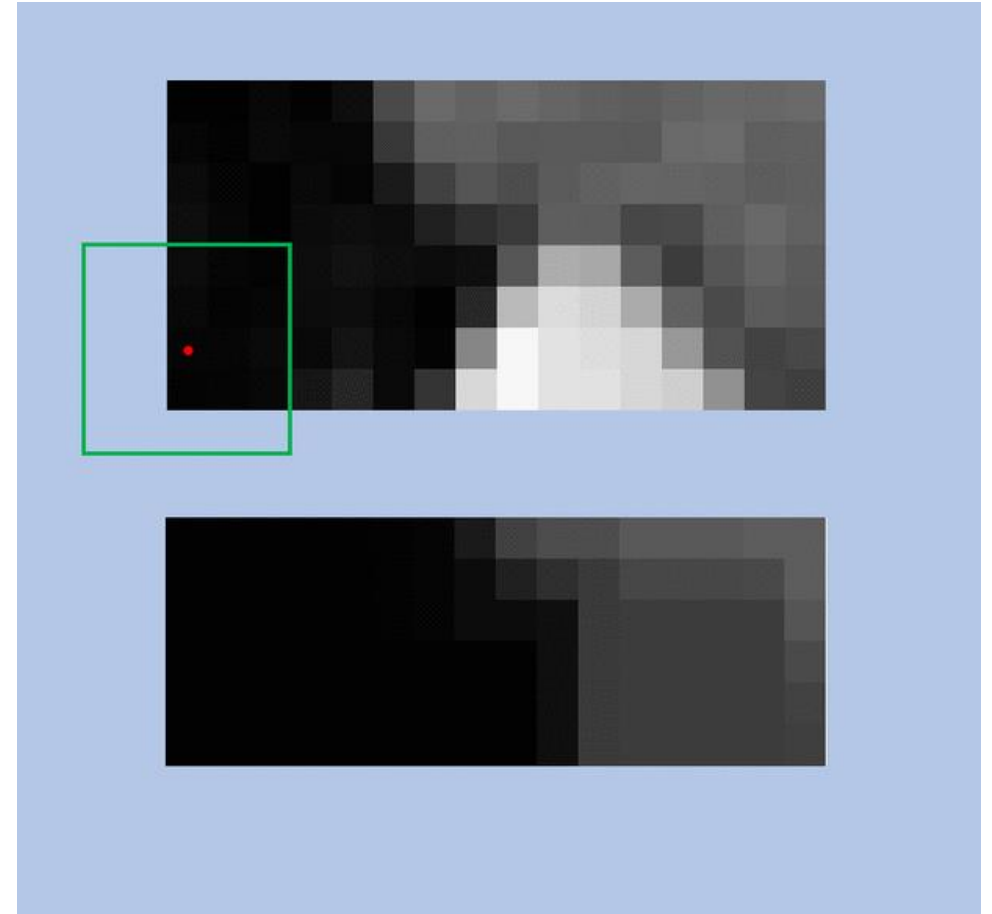


$$(I \ominus H)(u, v) = \min_{(i,j) \in H} \{I(u+i, v+j) - H(i, j)\}$$

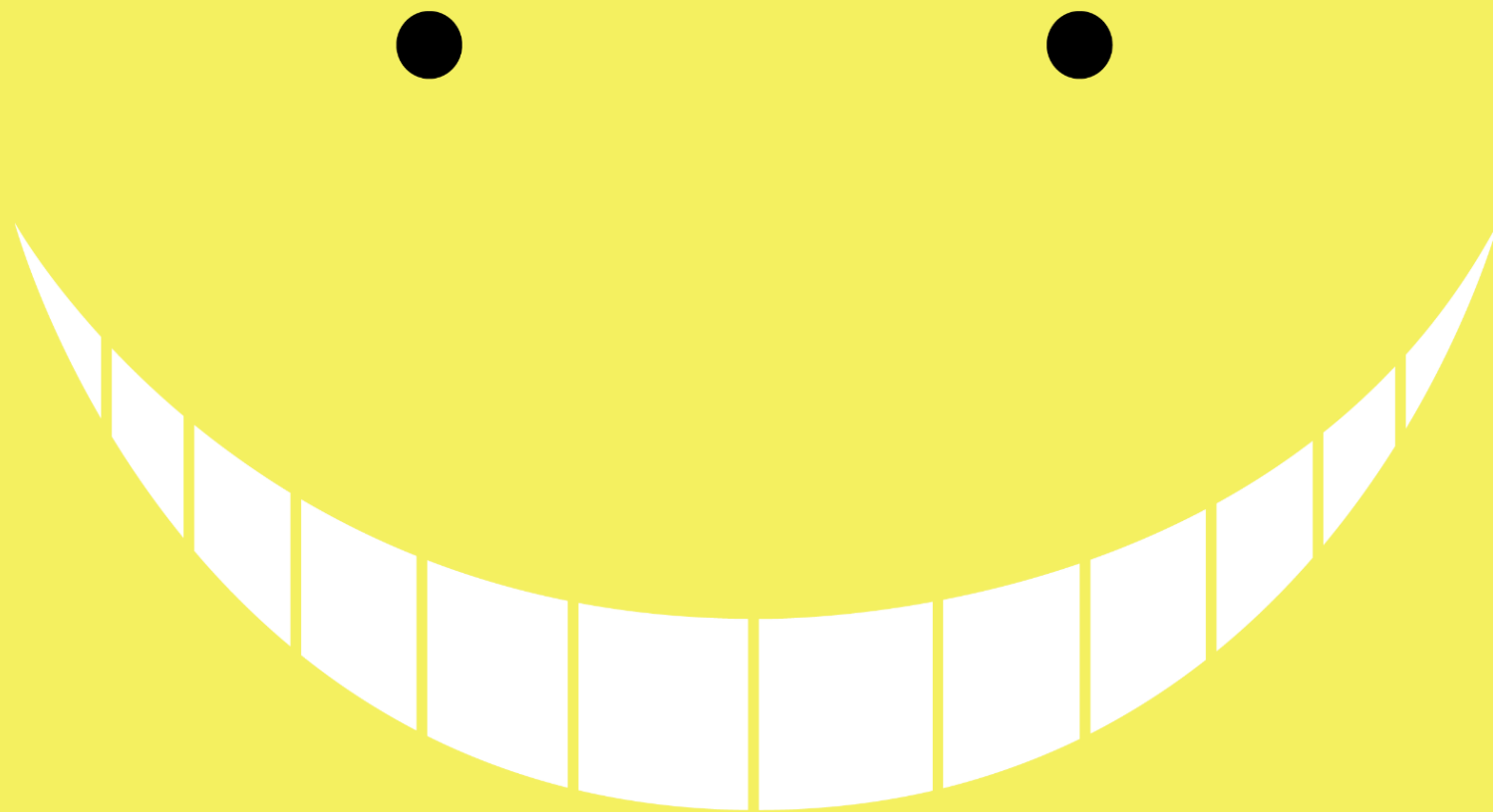
Grayscale morphology



Dilation



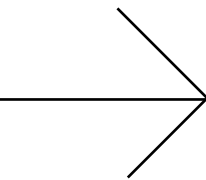
Erosion



The difference between the novice and the master is that the **master has failed more times** than the **novice has tried**.

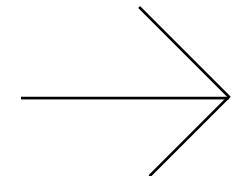
Evaluating Experimental Results using the **Confusion Matrix**: Principles and Practical Insights

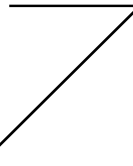
Understanding classification model performance through key metrics



Session Agenda

- Understanding the Confusion Matrix
- Key Evaluation Metrics Derived From the Confusion Matrix
- Practical Steps to Analyze Experimental Results
- Advanced Topics and Best Practices





Understanding the Confusion Matrix

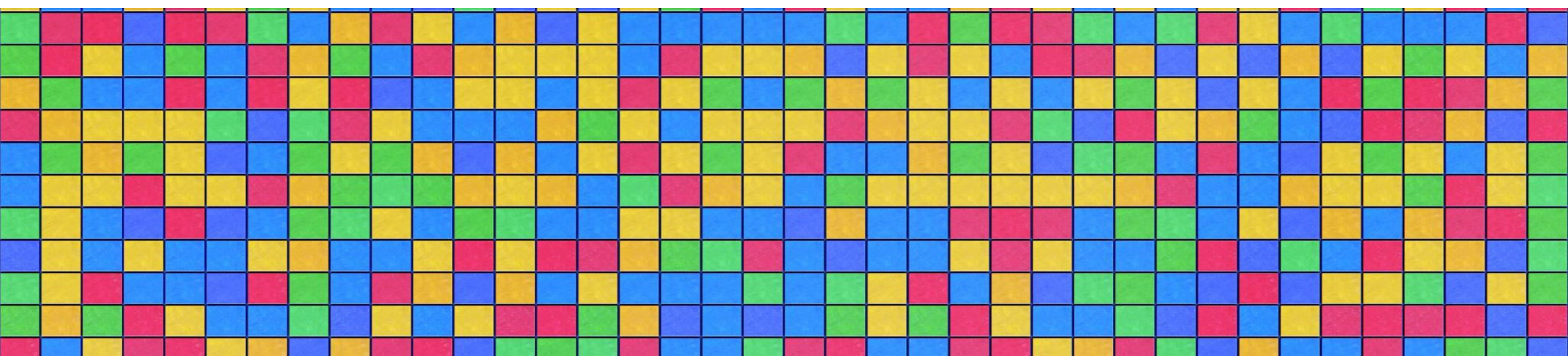
Definition and Structure of the Confusion Matrix

Confusion Matrix Overview

A confusion matrix is a table comparing predicted labels with actual labels in classification tasks.

Binary Classification Matrix

In binary classification, the confusion matrix is a 2x2 grid showing counts of true positives, true negatives, false positives, and false negatives.



Common Terms:

True Positives

False Positives

True Negatives

False Negatives

True Positives (TP)

True Positives are instances correctly predicted as positive, indicating accurate detection of positive cases.

False Positives (FP)

False Positives are incorrect positive predictions, representing errors where negative instances are wrongly labeled positive.

True Negatives (TN)

True Negatives are instances correctly predicted as negative, reflecting accurate identification of negative cases.

False Negatives (FN)

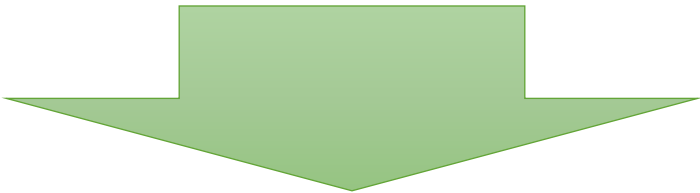
False Negatives are missed positive instances, indicating failures to identify positive cases correctly.



Binary Classification

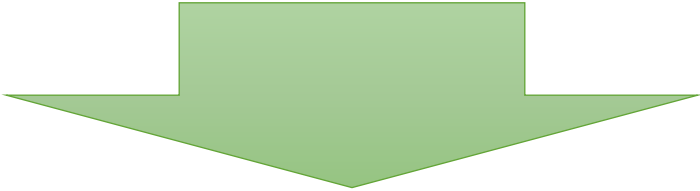


#	1	2	3	4	5	6	7	8	9	10
Actual classification	Dog	Dog	Dog	Dog	Cat	Cat	Cat	Cat	Cat	Cat



#	1	2	3	4	5	6	7	8	9	10
Actual classification	1	1	1	1	0	0	0	0	0	0

#	1	2	3	4	5	6	7	8	9	10
Actual classification	1	1	1	1	0	0	0	0	0	0



#	1	2	3	4	5	6	7	8	9	10
Actual classification	1	1	1	1	0	0	0	0	0	0
Predicted classification	0	1	1	1	1	1	0	0	0	0



#	1	2	3	4	5	6	7	8	9	10
Actual classification	1	1	1	1	0	0	0	0	0	0
Predicted classification	0	1	1	1	1	1	0	0	0	0
Result	FN	TP	TP	TP	FP	FP	TN	TN	TN	TN



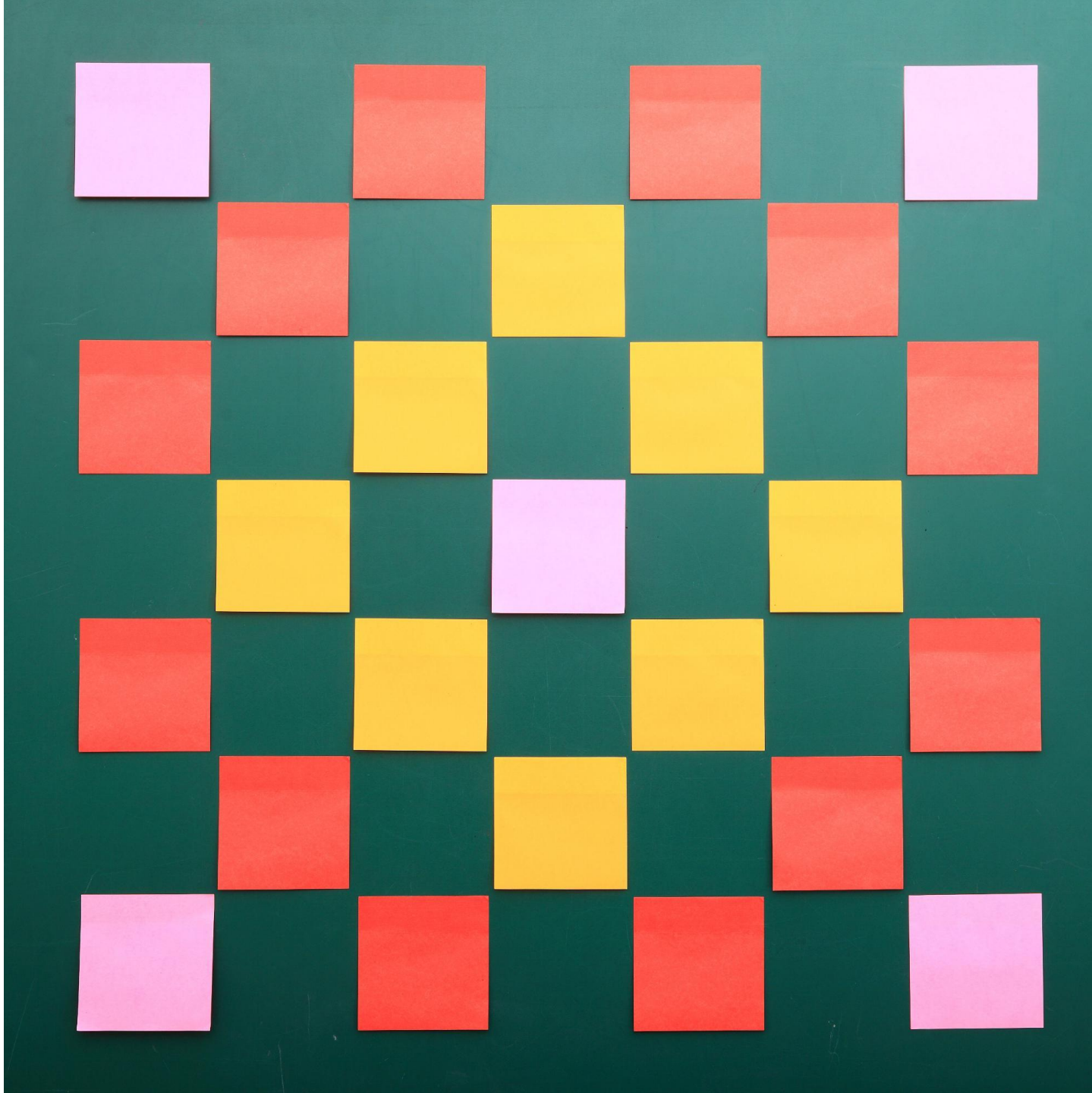
Visual Representation and Interpretation

Confusion Matrix Structure

The confusion matrix is displayed as a grid with actual classes on one axis and predicted classes on the other axis.

Model Performance Insights

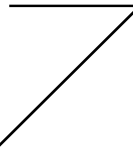
This visual format helps quickly identify areas where the model performs well and highlights the types of errors made.



#	1	2	3	4	5	6	7	8	9	10
Actual classification	1	1	1	1	0	0	0	0	0	0
Predicted classification	0	1	1	1	1	1	0	0	0	0
Result	FN	TP	TP	TP	FP	FP	TN	TN	TN	TN

		Predicted condition	
		Positive (PP)	Negative (PN)
Actual condition	Positive (P)	True Positive (TP)	False Negative (FN)
	Negative (N)	False Positive (FP)	True Negative (TN)

		Predicted condition	
		Total = 10	
Actual condition	Positive (P) = 4	Positive (PP) = 5 3	Negative (PN) = 5 1
	Negative (N) = 6	2	4



Key Evaluation Metrics

Derived From the Confusion Matrix



Calculating Accuracy, Precision, Recall, and F1-Score

Accuracy Overview

Accuracy measures the overall correctness of a model's predictions using true positives and true negatives.

Precision Defined

Precision measures the correctness of positive predictions by calculating **true positives over predicted positives**.

Recall Explained

Recall measures **how many actual positive cases are correctly identified** by the model.

F1-Score Balance

F1-score **balances precision and recall**, providing a single metric for model performance evaluation.

		Predicted condition	
		Positive (PP)	Negative (PN)
Actual condition	Total		
	Positive (P)	True Positive (TP)	False Negative (FN)
	Negative (N)	False Positive (FP)	True Negative (TN)

		Predicted condition	
		Positive (PP) = 5	Negative (PN) = 5
Actual condition	Total = 10		
	Positive (P) = 4	3	1
	Negative (N) = 6	2	4

$$\text{Accuracy (ACC)} = \frac{TP + TN}{P + N}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Fall-out (False Alarm)} = \frac{FP}{N} = \frac{FP}{FP + TN}$$

$$\text{Recall} = \frac{TP}{P} = \frac{TP}{TP + FN}$$

		Predicted condition			
		Total	Positive (PP)	Negative (PN)	
Actual condition	Positive (P)	True Positive (TP)	False Negative (FN)	$TPR = \frac{TP}{P}$	$FNR = \frac{FN}{N}$
	Negative (N)	False Positive (FP)	True Negative (TN)	$FPR = \frac{FP}{P}$	$TNR = \frac{TN}{N}$

True Positive Rate = Hit Rate

False Negative Rate = Miss Rate

False Positive Rate = False Alarm

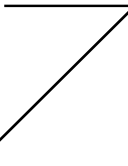
TPR = True Positive Rate

TNR = True Negative Rate

FPR = False Positive Rate **Error Type 1**

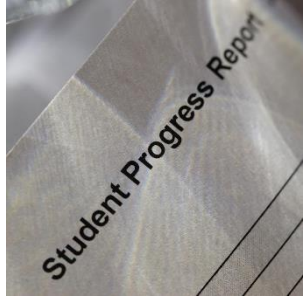
FNR = False Negative Rate **Error Type 2**

$$F_1 = 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$



Practical Steps to Analyze Experimental Results

Constructing and Populating the Confusion Matrix From Results



Collect True and Predicted Labels

Gather true labels and predicted labels from experiment results for analysis.



Tabulate Counts in Matrix

Organize counts of label matches and mismatches into a confusion matrix format.



Ensure Accurate Construction

Accurately building the matrix is critical for reliable analysis and interpretation.



Interpreting Results in Context of Experimental Objectives

Contextual Interpretation

Results must be interpreted based on the experiment's specific objectives to ensure relevance and accuracy.

Minimizing False Negatives

Reducing false negatives is **crucial when missing a positive result has severe consequences.**

Minimizing False Positives

Lowering false positives is important where incorrect positive results could cause harm or **waste resources.**

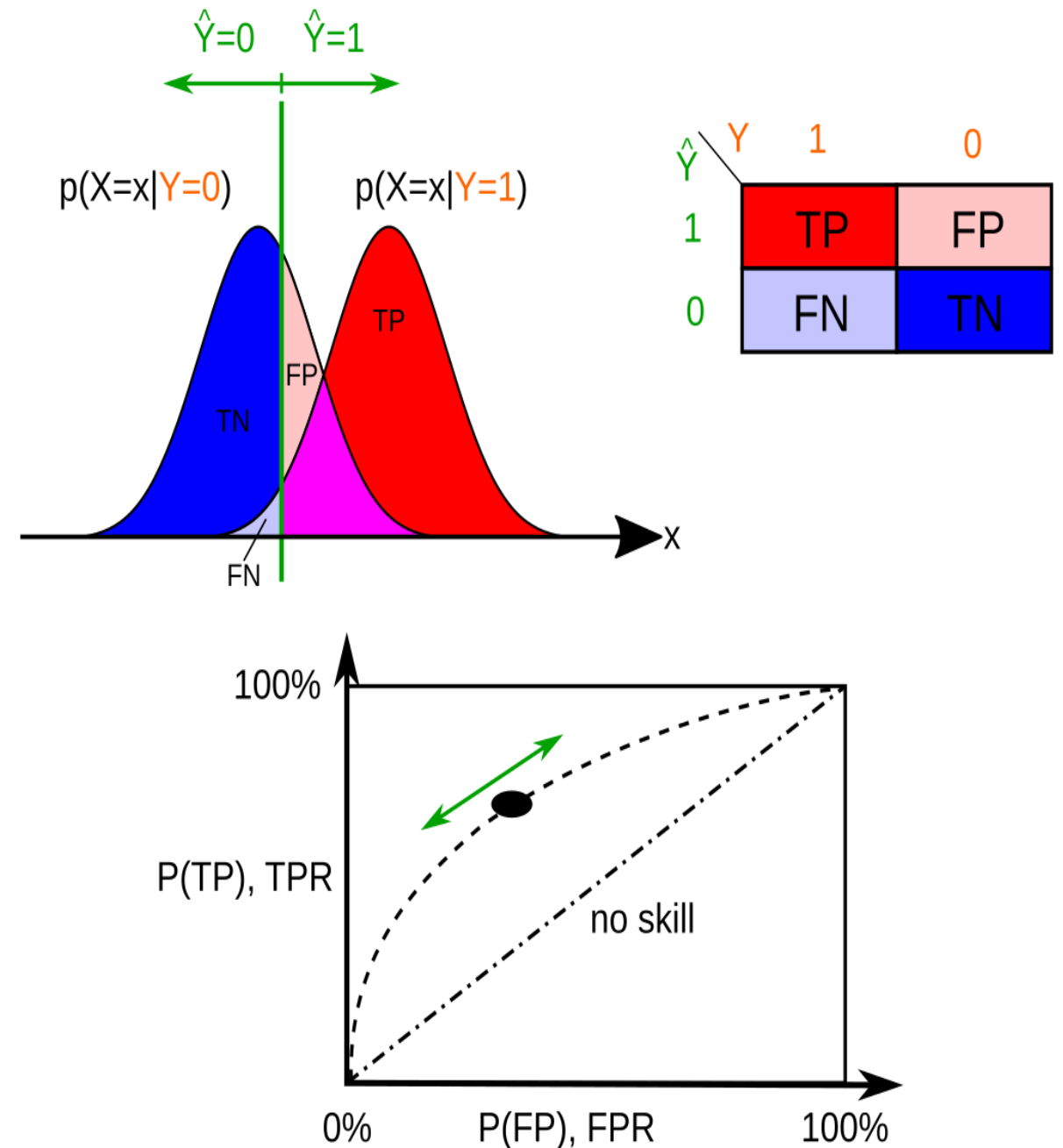
Guiding Improvements

Understanding errors guides meaningful conclusions and improvements in experimental methods.

Receiver Operating Characteristic (ROC) curve

A **receiver operating characteristic curve**, or **ROC curve**, is a graphical plot that illustrates the performance of a **binary classifier** at **varying threshold values**.

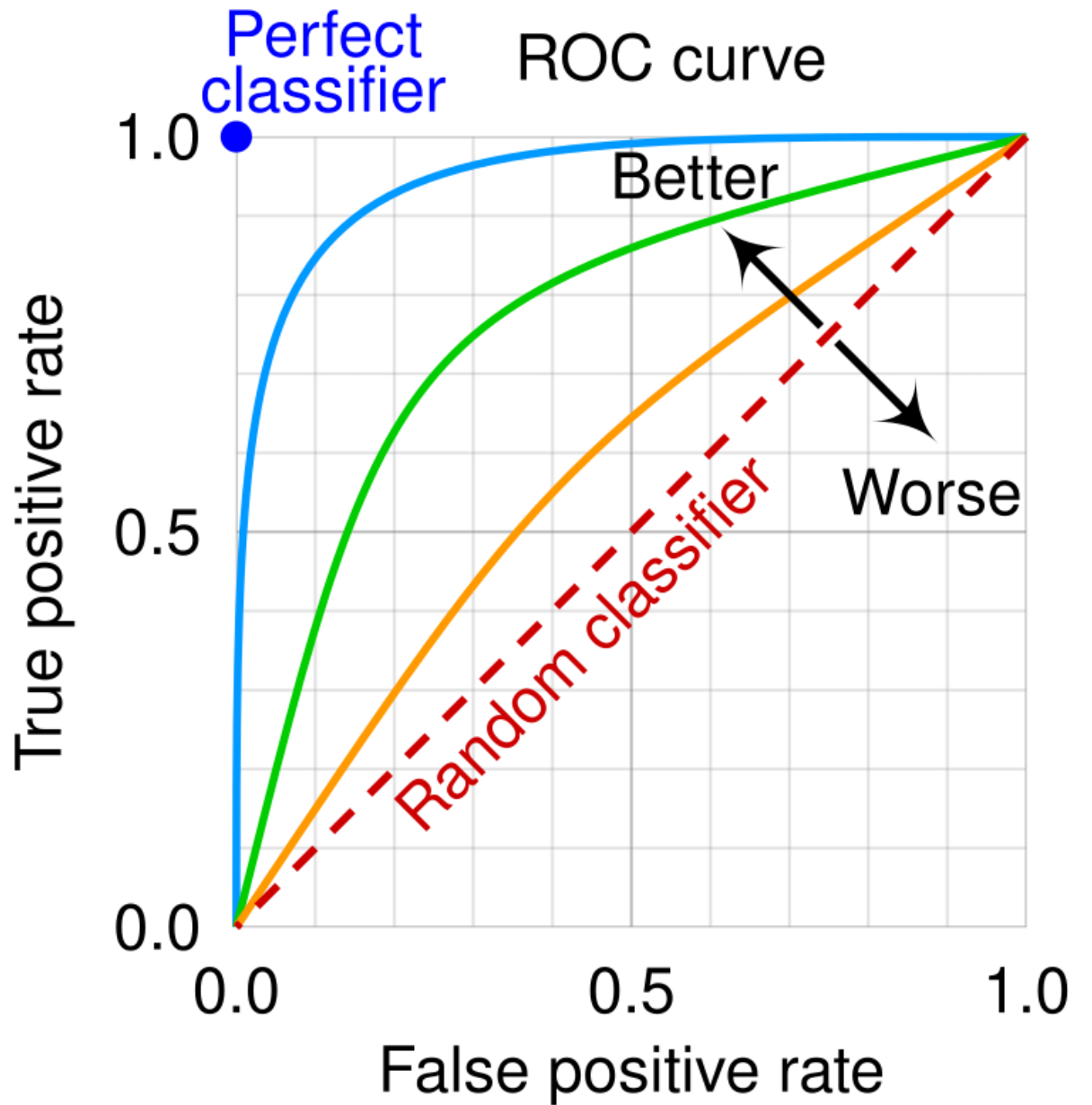
ROC analysis is commonly applied in the assessment of diagnostic test performance in clinical epidemiology.

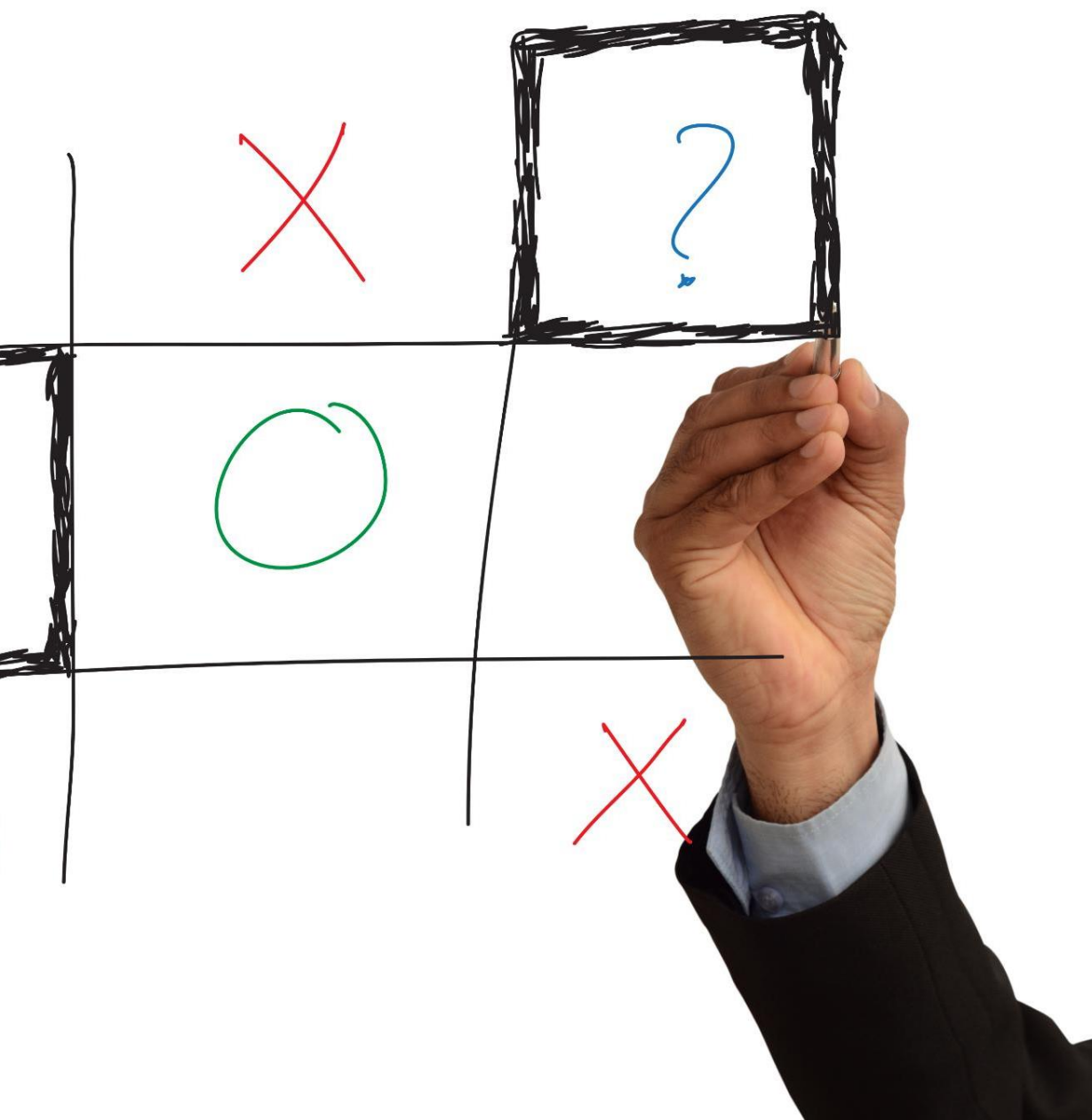


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Practice with Confusion Matrix

Building Confusion Matrices

Regular practice with confusion matrices enhances understanding of classification problems for students and professionals alike.

Visualizing Model Performance

Confusion matrices display true positives, false positives, true negatives, and false negatives to visualize model performance clearly.

Analyzing Strengths and Weaknesses

Examining confusion matrices helps identify strengths and weaknesses in predictive models, guiding further improvements.

Conclusion

Confusion Matrix Importance

The confusion matrix is essential for assessing the performance of classification models accurately and comprehensively.

Derived Metrics

Metrics such as accuracy, precision, recall, and F1-score are derived from the confusion matrix to evaluate models.

Practical Application

Understanding and using the confusion matrix enables informed decisions and improvements in experimental outcomes.

Workshop (งานกลุ่ม)

1. หาโจทย์การแบ่งส่วน (Segmentation) วัตถุ (Object) ออกจากพื้นหลัง (Background)
2. หา Dataset อย่างน้อย 50 ภาพ และ Ground truth (ถ้าไม่มี ให้สร้างขึ้นมาเอง)
3. ออกแบบและสร้าง Algorithm เพื่อแยกวัตถุออกจากพื้นหลัง โดยกำหนดให้วัตถุที่ต้องการเป็น Positive Class และพื้นหลังเป็น Negative Class
4. ทดลองใช้ Morphology algorithm เพื่อปรับปรุงผลลัพธ์กระบวนการ Segmentation
5. วัดประสิทธิภาพการทดลอง (Evaluation) โดยให้แสดงผลลัพธ์ Confusion Matrix และ ROC Curve พร้อมทั้งอภิปรายผลลัพธ์ที่ได้
6. นำเสนออาจารย์

<https://www.geeksforgeeks.org/computer-vision/image-segmentation-approaches-and-techniques-in-computer-vision/>